

LOCALIZATION OF RELATIVE VIRTUAL CLASSES (PRELIMINARY DRAFT)

THOMAS GRABER AND ISAAC MARTIN

ABSTRACT. We prove a generalization of the classic virtual localization formula to virtual classes defined over an algebraic stack and show that our formula holds for the moduli space of punctured stable logarithmic maps. This is a preliminary draft.

CONTENTS

1. Introduction	2
2. The fixed stack and its deformation theory	6
2.1. The Romagny fixed stack	6
2.2. Torus actions and algebraicity of the fixed stack	8
2.3. Deformation theory of the fixed stack	10
2.4. The fixed stack need not be algebraic	13
3. The residue map for Artin stacks	15
3.1. Euler classes of vector bundle stacks	17
3.2. The residue map	18
4. Localization	26
4.1. The fixed locus of the base change	28
4.2. Residues, base change and compatible triples	29
4.3. The Localization Theorem	33
4.4. Comparison to the absolute case	34
5. Punctured Stable Maps	35
References	40

1. INTRODUCTION

Let M be a smooth compact manifold equipped with the action of a torus T and let $i : M^T \rightarrow M$ denote the inclusion of the fixed locus. Atiyah and Bott introduced torus localization [AB84] as a means of relating the equivariant cohomology of M and M^T , thereby providing a way to perform cohomological computations on M via the simpler cohomology of M^T . Their setup consists of two parts. First, their main localization theorem states that the pushforward i_* on T -equivariant cohomology becomes an isomorphism after localizing at a certain multiplicative set:

$$(1) \quad i_* : A_*^T(M^T)_{\text{loc}} \xrightarrow{\sim} A_*^T(M)_{\text{loc}}$$

Second, the Bott residue formula provides an explicit description of both the image of the pushforward i_* and its inverse, giving a formula for expressing classes in $\alpha \in A_*^T(M)$ in terms of classes in $A_*^T(M^T)$:

$$(2) \quad \alpha = \sum_{F \subseteq M^T} i_* \left(\frac{\alpha|_F}{e_T(N_{F/M})} \right)$$

where the sum taken over the connected components of M^T and $e_T(N_{F/M})$ is T -equivariant Euler class of the normal bundle of F in M .

Progress in torus localization over the 1990s can be viewed as steady generalizations of the theory to increasingly singular settings, spurred onward by desired applications in enumerative geometry and mirror symmetry. Kontsevich applied Bott's formula to the smooth moduli stack $\overline{M}_{0,k}(\mathbb{P}^n, d)$ to test enumerative mirror symmetry numerically [Kon95]. Edidin and Graham proved (1) for arbitrary schemes with torus actions and proved (2) for smooth schemes [EG98] in 1997. Then in 1999, the first author and Pandharipande extended the localization theorem to virtually smooth spaces and proved a version of the Bott residue formula for virtual classes [GP99]. This opened the door to localization computations on much wider class of moduli spaces, including Li's moduli space of relative stable maps [GV05] and Kim's logarithmic variant the moduli space of stable logarithmic maps to expansions of a smooth pair (X, D) [MR19].

However, the virtual localization theorem of Graber-Pandharipande does not apply directly to the moduli space of stable logarithmic maps – the localization computations of [GV05; MR19] instead pass to moduli spaces of maps to expansions. To run localization on M over a field k , they require that

- (1) k is algebraically closed
- (2) M embeds in an ambient smooth space
- (3) M is virtually smooth over k , i.e. there exists an *absolute* perfect obstruction theory $\phi : E \rightarrow L_{M/k}$.

asdf For a fs logarithmic scheme X and a choice of discrete data β , the moduli space $\mathcal{M}(X, \beta)$ of basic stable logarithmic maps to X is only virtually smooth *relative* to the moduli space of basic pre-stable log curves $\mathfrak{M}$, that is, the virtual fundamental class of $\mathcal{M}(X, \beta)$ is defined via a relative perfect obstruction theory $\phi : E \rightarrow L_{\mathcal{M}(X, \beta)/\mathfrak{M}}$. While hypotheses (1) and (2) above have been removed in the intervening years since 1999 — (1) for trivial cosections in [CKL17] and then both (1) and (2) for all cases in [Ara+25] — the absolute virtual smoothness hypothesis remains.

The problem is as follows. Let T be an algebraic torus over a field k , $\pi : M \rightarrow \mathcal{Y}$ a T -equivariant morphism from a Deligne-Mumford stack to an algebraic stack over k and $\phi : E \rightarrow L_{M/\mathcal{Y}}$ a T -equivariant perfect obstruction theory for π in the sense of [BF97]. The Romagny fixed stack construction is functorial [Rom05] and gives us a commutative square

$$(3) \quad \begin{array}{ccc} M^T & \xrightarrow{i} & M \\ \pi^T \downarrow & & \downarrow \pi \\ \mathcal{Y}^T & \xrightarrow{\eta} & \mathcal{Y} \end{array} .$$

If we additionally take M and $\mathcal{Y}$ both to be finite type over k , we can access the T -equivariant Chow groups for M and $\mathcal{Y}$ by taking $A_*^T(-) := A_*([-/T])$ [Kre99], and from the obstruction theory E we obtain virtual classes in $A_*^T(M)$ via Manolache's virtual pullback. The localization isomorphism exists for Deligne-Mumford stacks (Proposition 2.7), so $i_* : A_*^T(M^T) \rightarrow A_*^T(M)$ has an inverse which we denote by $\text{Res}_{M^T/M}$. To complete the picture, we need to specify how virtual classes in M^T relate to those in M , and it is here that we run into problems.

The difficulties are summarized in the following diagram:

$$(4) \quad \begin{array}{ccc} A_*^T(M^T)_{\text{loc}} & \xleftarrow{\text{Res}_{M^T/M}} & A_*^T(M)_{\text{loc}} \\ \uparrow (a) & & \uparrow \pi^! \\ \boxed{?} & \xleftarrow{(c)} & A_*^T(\mathcal{Y})_{\text{loc}} \\ (b) & & \end{array}$$

In the absolute situation, taking the fixed part of E over M^T gives an an obstruction theory for M^T over $(\text{Spec } k)^T = \text{Spec } k$, thereby defining a virtual fundamental classes for M^T and M over the same base. In our situation, virtual classes for M^T are pulled back from $\mathcal{Y}^T$, and in order to maintain compatability with $\pi^!$, the perfect obstruction theory for $\pi^T : M^T \rightarrow \mathcal{Y}^T$ must be canonical in some way. This is done in §2.3, yielding a virtual pullback $(\pi^T)^!$ for (a).

There is also the matter of defining a sensible module of invariants for $\mathcal{Y}^T$. The Romagny fixed stack may not be algebraic, and even when it is, it is rarely finite

type over k . Choosing a replacement $\mathcal{Z}$ for $\mathcal{Y}^T$ and suitable set at which to localize is §3.1 and yields $A_*^T(\mathcal{Z})_{\text{loc}}$ in position (b).

Finally, the sensible map for position (c) above would be the inverse of the localization isomorphism. Here the theory breaks down completely. Even after choosing a suitable $\mathcal{Z} \subseteq \mathcal{Y}^T$ in order to define the Chow groups $A_*^T(\mathcal{Z})$, $\eta|_{\mathcal{Z}} : \mathcal{Z} \rightarrow \mathcal{Y}$ often fails to be a monomorphism and hence η_* is almost never an isomorphism, localization or not. Instead, we construct a map $\text{Res}_{\mathcal{Z}/\mathcal{Y}} : A_*^T(\mathcal{Y})_{\text{loc}} \rightarrow A_*^T(\mathcal{Z})_{\text{loc}}$ directly in §3.2.

This gives us a diagram

$$\begin{array}{ccc} A_*^T(M^T)_{\text{loc}} & \xleftarrow{\text{Res}_{M^T/M}} & A_*^T(M)_{\text{loc}} \\ (\pi^T)^! \uparrow & & \uparrow \pi^! \\ A_*^T(\mathcal{Z})_{\text{loc}} & \xleftarrow{\text{Res}_{\mathcal{Z}/\mathcal{Y}}} & A_*^T(\mathcal{Y})_{\text{loc}} \end{array}$$

whose commutativity is the content of our relative virtual localization theorem (proven as Theorem 4.8).

Theorem A. *For every $\beta \in A_*^T(\mathcal{Y})$,*

$$(\pi^T)^!_{E^f} (\text{Res}_{\mathcal{Z}/\mathcal{Y}}(\beta)) = e_T(N^{\text{vir}}) \cap \text{Res}_{M^T/M} (\pi_E^!(\beta)) \quad \text{in } A_*^T(M^T)_{\text{loc}},$$

and if $\mathcal{Y}$ is pure-dimensional, then

$$(\pi^T)^!_{E^f} (\alpha) = e_T(N^{\text{vir}}) \cap \text{Res}_{M^T/\widetilde{M}} ([M]^{\text{vir}})$$

for $\alpha = \text{Res}_{\mathcal{Z}/\mathcal{Y}}([\mathcal{Y}])$.

Here $\mathcal{Z} \subseteq \mathcal{Y}^T$ is any open and closed substack of finite type containing the image of M^T , $\widetilde{M} = M \times_{\mathcal{Y}} \mathcal{Z}$ is the base change, E^f is the fixed part of $E|_{M^T}$, which is a perfect obstruction theory for $\pi^T : M^T \rightarrow \mathcal{Y}^T$ by Lemma 2.12, N^{vir} is the moving part of $E^\vee|_{M^T}$, and $\text{Res}_{M^T/\widetilde{M}}$ is the residue map of the Deligne-Mumford stack $\widetilde{M}$, which is the inverse of the localization isomorphism. When $\mathcal{Y}$ is a point, $\text{Res}_{\mathcal{Z}/\mathcal{Y}}$ is the identity and the theorem reduces to the formula of [GP99] (Corollary 4.12).

The contributions of this paper are the following.

- (1) A canonical perfect obstruction theory $(E|_{M^T})^{\text{fix}}$ for $M^T \rightarrow \mathcal{Y}^T$, compatible with E (Lemma 2.12), together with the properties of the Romagny fixed stack of an algebraic stack needed to work with it (Proposition 2.7).
- (2) A residue map $\text{Res}_{\mathcal{Z}/\mathcal{Y}} : A_*^T(\mathcal{Y}) \rightarrow A_*^T(\mathcal{Z})_{\text{loc}}$ for an algebraic stack $\mathcal{Y}$, replacing the inverse of η_* , which does not exist (Definition 3.12). It is defined through a choice of embedding data and shown to be independent of it (Lemma 3.6).

- (3) The localization formula (Theorem 4.8) and its specialization to the classical virtual localization formula (Corollary 4.12).
- (4) Verification that the bases of punctured logarithmic maps satisfy the standing hypotheses of the formula (Proposition 5.4).

Section 2 deals with properties of the Romagny fixed stack and the morphism $\pi^T : M^T \rightarrow \mathcal{Y}^T$: §2.1 reviews the definition of the Romagny fixed stack, §2.2 establishes properties we need in the case of torus actions and §2.3 studies the deformation theory of $\mathcal{Y}^T$ to resolve issue (a) in (4). Section 3 concerns the construction of the residue map for algebraic stacks: §3.1 discusses Euler classes of vector bundle stacks and sets up a reasonable localized module of invariants for $\mathcal{Z} \subseteq \mathcal{Y}$ and §3.2 constructs the residue map. The main result is proven in Section 4, and §4.4 shows that it recovers the absolute virtual localization theorem when $\mathcal{Y} = \operatorname{Spec} k$. Section 5 proves that the necessary hypotheses for the main theorem are satisfied by the moduli space of basic punctured stable logarithmic maps, the main application we have in mind.

This is a preliminary draft. Theorem 4.8 and Corollary 4.12 are proven in full under Convention 1.1 and the existence of the resolutions (a) and (b) of Remark 4.3. Section 5 shows that the bases of punctured logarithmic maps satisfy Convention 1.1 (Proposition 5.3). What is missing is the verification that resolution (a), a two-term resolution of $\tau_{\geq -1} L_\eta$ by moving locally free sheaves, exists in that setting. We have not yet determined in what generality it exists, and the end of Section 5 is accordingly incomplete. The draft has also not been thoroughly proofread and is likely accordingly rough around the edges. We make it public now because it is used in the recent preprint [SG26]. The version cited there may differ from this one in numbering.

Below is a list of conventions that apply throughout the paper.

Convention 1.1.

- (1) *Ground field and torus.* We work over a field k , T is an algebraic torus of rank r over k with character lattice Λ .
- (2) *Stacks.* All algebraic stacks are locally of finite type over k , with affine stabilizers and quasi-compact separated diagonal, unless stated otherwise. Plain letters (M, X) denote Deligne-Mumford stacks and schemes, calligraphic letters $(\mathcal{X}, \mathcal{Y}, \mathcal{H})$ algebraic stacks, fraktur letter $\mathfrak{C}, \mathfrak{E}$ vector bundle and cone stacks and later $\mathfrak{M}$ in Section 5 for the bases of logarithmic curves and maps.

- (3) *Chow groups.* We use Kresch's Chow groups [Kre99, Def. 2.1.11] with $\mathbb{Q}$ -coefficients, and hence only consider Chow groups for algebraic stacks finite type over k (i.e. stacks $\mathcal{Y}$ which are quasi-compact in addition to the hypotheses above). For such a stack $\mathcal{Y}$ with a T action, $A_*^T(\mathcal{Y}) := A_*([\mathcal{Y}/T])$ and $A_T^*(\mathcal{Y})$ is the ring of operators on it generated by Chern classes of T -equivariant vector bundles [Kre99, Thm. 2.1.12]. The localized Chow groups $A_*^T(-)_{\text{loc}}$ are those of Section 3.1.
- (4) *Fixed stack.* For a T -stack $\mathcal{Y}$, $\mathcal{Y}^T$ is Romagny's fixed stack [Rom05] with its morphism $\eta : \mathcal{Y}^T \rightarrow \mathcal{Y}$. Its objects are described in Remark 2.2 and its properties are collected in Proposition 2.7. The stack $\mathcal{Y}^T$ carries the trivial T -action.

2. THE FIXED STACK AND ITS DEFORMATION THEORY

In this section we recall Romagny's fixed stack $\mathcal{Y}^T$ of an algebraic stack with a torus action and establish certain facts about it used throughout the rest of the paper. Proposition 2.7 collects the properties of $\eta : \mathcal{Y}^T \rightarrow \mathcal{Y}$ (algebraicity, representability and the localization isomorphism for Deligne-Mumford stacks) that Kresch's Chow groups and Manolache's virtual pullback require. Lemma 2.11 shows that $\tau_{\geq -1} L_{\mathcal{Y}^T/\mathcal{Y}}$ has no fixed part, and Lemma 2.12 deduces from it that the fixed part of a perfect obstruction theory for $M \rightarrow \mathcal{Y}$ is a perfect obstruction theory for $M^T \rightarrow \mathcal{Y}^T$. Identifying the Romagny fixed stack with the classical truncation of the derived homotopy fixed stack $\mathcal{Y}^{hT}$ of [Ara+25] streamlines our work considerably (see Remark 2.6). Under Convention 1.1 the fixed stack is always algebraic, but section 2.4 records a way in which this fails over a base which is not a field.

2.1. The Romagny fixed stack. The group actions on stacks we consider are those of Romagny [Rom05, Def. 2.1], and by [Rom05, Prop. 1.5] we can always take the necessary action diagrams to strictly 1-commute. An algebraic stack with an action of G is called a G -stack. In this setup the fixed stack is defined as follows:

Definition 2.1. ([Rom05, Def. 2.3]) Let G be a group algebraic space over a scheme S and let $\mathcal{Y}$ be a G -stack over S . For an S -stack $\mathcal{X}$, denote by $\iota(\mathcal{X})$ the G -stack over S obtained by endowing $\mathcal{X}$ with the trivial G -action. The *stack of fixed points* $\mathcal{Y}^G$ is a stack that 2-represents the functor

$$\mathcal{H}om_{G\text{-St}}(\iota(-), \mathcal{Y}),$$

and there is a canonical morphism $\eta : \mathcal{Y}^G \rightarrow \mathcal{Y}$ which can be considered a G -equivariant morphism by endowing $\mathcal{Y}^G$ with the trivial G -action.

The next two remarks give different descriptions of the fixed stack. The first is the one we will use most, and the second is used to prove algebraicity and separatedness of $\mathcal{Y}^T \rightarrow \mathcal{Y}$ in Section 2.2.

Remark 2.2. An object of $\mathcal{Y}^G(U)$ consists of a G -equivariant morphism $(x, \alpha) : \iota(U) \rightarrow \mathcal{Y}$ fitting into a diagram

$$\begin{array}{ccc} G \times U & \xrightarrow{\text{pr}_2} & U \\ \text{id}_G \times x \downarrow & \nearrow \alpha & \downarrow x \\ G \times \mathcal{Y} & \xrightarrow{\mu} & \mathcal{Y} \end{array}$$

The morphism x is an object of $\mathcal{Y}$ the 2-commutativity of the diagram means that for all $g \in G(U)$ we get an isomorphism $\alpha_g : g \cdot x \xrightarrow{\sim} x$. Similarly, given a pair $(x, \{\alpha_g\}_{g \in G(U)})$ consisting of an object $x \in \mathcal{Y}(U)$ and a collection of isomorphisms $\alpha_g : gx \xrightarrow{\sim} x$, we obtain a G -equivariant map $\iota(U) \rightarrow \mathcal{Y}$. Thus the objects of $\mathcal{Y}^G(U)$ are precisely pairs $(x, \{\alpha_g\}_{g \in G(U)})$ where $x \in \mathcal{Y}(U)$ and the $\alpha_g : g \cdot x \xrightarrow{\sim} x$ are isomorphisms satisfying the cocycle condition $\alpha_{gh} = \alpha_g \circ g \cdot \alpha_h$ pictured by

$$(5) \quad \begin{array}{ccc} & gx & \\ g \cdot \alpha_h \nearrow & & \searrow \alpha_g \\ (gh)x & \xrightarrow{\alpha_{gh}} & x \end{array}$$

for all $g, h \in G(U)$. The morphism $\eta : \mathcal{Y}^G \rightarrow \mathcal{Y}$ simply forgets the data of α .

In particular, when G acts trivially on $\mathcal{Y}$, $gx = x$ for all $x \in \mathcal{Y}(U)$ and the cocycle condition reduces to the requirement that $\alpha_{gh} = \alpha_g \circ \alpha_h$; i.e. that $\alpha : G \rightarrow \underline{\text{Aut}}(x)$ is a group homomorphism. Thus, for the trivial action, the fiber of $x \in \mathcal{Y}(U)$ over η is the sheaf $\underline{\text{Hom}}_{S-gp}(G, \text{Aut}(x))$.

Remark 2.3. The stack $\mathcal{Y}^G$ also admits a description via something called the *universal stabilizer* $\text{St}_{\mathcal{Y}, G}$ of the action of G on $\mathcal{Y}$ [Rom22, Section 4.1]. It is the fiber product

$$(6) \quad \begin{array}{ccc} \text{St}_{\mathcal{Y}, G} & \longrightarrow & \mathcal{Y} \\ \downarrow & & \downarrow \Delta \\ G \times_S \mathcal{Y} & \xrightarrow{\text{act} \times \text{pr}_2} & \mathcal{Y} \times \mathcal{Y} \end{array} .$$

Over a point $x : U \rightarrow \mathcal{Y}$ its objects are

$$\text{St}_{\mathcal{Y}, G}(U) = \{(g, \alpha) \mid g \in G(U), \alpha : gx \xrightarrow{\sim} x\},$$

with multiplication law $(g, \alpha) \cdot (h, \beta) = (gh, \alpha \circ g\beta)$ and identity element $(1, \text{id}_x)$. The leftmost vertical arrow in diagram (6) is given by $(g, \alpha) \mapsto g$ over $x : U \rightarrow \mathcal{Y}$

and is a morphism of $\mathcal{Y}$ -group stacks. Its kernel is the inertia stack $I_{\mathcal{Y}} := \mathcal{Y} \times_{\mathcal{Y} \times \mathcal{Y}} \mathcal{Y}$, and thus we get an exact sequence

$$(7) \quad 1 \rightarrow I_{\mathcal{Y}/S} \rightarrow \mathrm{St}_{\mathcal{Y},G} \rightarrow G \times_S \mathcal{Y}$$

where over $x : U \rightarrow \mathcal{Y}$ the map $I_{\mathcal{Y}/S} \rightarrow \mathrm{St}_{\mathcal{Y},G}$ is

$$(\alpha : x \xrightarrow{\sim} x) \mapsto (1, \alpha).$$

We call the cokernel of $I_{\mathcal{Y}/S}(x) \hookrightarrow \mathrm{St}_{\mathcal{Y},G}(x)$ the G -*stabilizer* of x . As an aside, the rightmost map in (7) admits a section over U if and only if x is in the image of $\eta(U) : \mathcal{Y}^G(U) \rightarrow \mathcal{Y}(U)$; indeed $\mathcal{Y}^G$ parameterizes group-theoretic sections of $\mathrm{St}_{\mathcal{Y},G} \rightarrow G \times_S \mathcal{Y}$ [Rom22, Lemma 4.1.2].

When there are many choices of α_g for a single $x \in \mathcal{Y}(U)$ we get multiple points in $\mathcal{Y}^G(U)$ mapping to a single point in $\mathcal{Y}$. This means that $\eta : \mathcal{Y}^G \rightarrow \mathcal{Y}$ is, in general, not a monomorphism.

Example 2.4. Let $\mathfrak{M}$ denote the stack of nodal curves of some fixed genus and endow it with the trivial action of an algebraic group G . Consider a scheme B . Specializing Remark 2.2 to this situation, we see that the points of $\mathfrak{M}^G(B)$ are given by pairs (x, α) where $x \in \mathfrak{M}(B)$ and $\alpha : G(B) \rightarrow \mathrm{Aut}(x)$ is a group homomorphism. The point x defines a family of nodal curves $E \rightarrow B$ over B by pulling back the universal curve over $\mathfrak{M}$, and α gives an automorphism of this family for every element of $G(B)$. Thus, $\mathfrak{M}^G$ parameterizes families of nodal curves with an action of G_B on E over B , and $\eta : \mathfrak{M}^G \rightarrow \mathfrak{M}$ is the map that forgets this G -action. Whenever there is such a family $E \rightarrow B$ with a nontrivial G_B -action on E , η will not be a monomorphism. The bases $\mathfrak{M}$ of Section 5 are of this kind.

Example 2.5. Let S be a scheme with a trivial $\mathbb{G}_m$ action, and endow $\mathcal{Y} = [S/\mathbb{G}_m] = B\mathbb{G}_m$ with the trivial $T = (\mathbb{G}_m)^n$ action. Every point in $x \in \mathcal{Y}$ is fixed by T , so a collection $\{\alpha_t\}$ satisfying the cocycle condition (5) is the same data as a homomorphism $\alpha : T \rightarrow \mathrm{Aut}_{\mathcal{Y}}(x) = \mathbb{G}_m$, i.e. a character of the torus. The fixed stack $\mathcal{Y}^T$ is therefore a countable union of copies of $\mathcal{Y}$:

$$\mathcal{Y}^T = \coprod_{\chi \in \mathbb{Z}^n} \mathcal{Y},$$

and the fiber of η at a point $x \in \mathcal{Y}$ is $\mathbb{Z}^n$. In particular, η is not a monomorphism and is not quasi-compact. The pushforward η_* , which the classical localization theorem inverts, is therefore not even defined; the residue map of Section 3 is its replacement.

2.2. Torus actions and algebraicity of the fixed stack. From now on $G = T$ is a torus over k . Kresch's Chow groups and Manolache's virtual pullback, on which everything from Section 3 onwards is built, require some control over the stack $\mathcal{Y}^T$

and the morphism $\eta : \mathcal{Y}^T \rightarrow \mathcal{Y}$. Many of the properties we need are proven in [Ara+25, Appendix A], so most of the work here is contained in translating their results for derived stacks to classical stacks.

We write $\mathcal{Y}^{hT}$ for the derived homotopy fixed stack [Ara+25, Def. A.14], $\varepsilon : \mathcal{Y}^{hT} \rightarrow \mathcal{Y}$ for its canonical map to $\mathcal{Y}$ [Ara+25, Rem. A.15] and $\tau : (\mathcal{Y}^{hT})_{\text{cl}} \rightarrow \mathcal{Y}^{hT}$ for the inclusion of its classical truncation.

Remark 2.6. Romagny's fixed stack $\mathcal{Y}^T$ is the classical truncation of the homotopy fixed stack $\mathcal{Y}^{hT}$ of [Ara+25, Def. A.14] and η factors as $\eta = \varepsilon \circ \tau$. Both are the stack of group-theoretic sections of $\text{St}_{\mathcal{Y}, T} \rightarrow T \times \mathcal{Y}$, the former on classical and the latter on derived test objects.

The following proposition collects the properties on η needed for the later sections.

Proposition 2.7 (Properties of the fixed stack). *Let T be a torus over k acting on an algebraic stack $\mathcal{Y}$ locally of finite type over k .*

- (a) *If $\mathcal{Y}$ has affine stabilizers and quasi-compact separated diagonal, then $\mathcal{Y}^T$ is an algebraic stack locally of finite type over k with affine stabilizers, and $\eta : \mathcal{Y}^T \rightarrow \mathcal{Y}$ is separated and representable by algebraic spaces. In particular η is of DM-type.*
- (b) *If $\mathcal{Y}$ is Deligne–Mumford, quasi-separated and has separated diagonal, then η is a closed immersion and $\mathcal{Y}^T$ is a quasi-separated Deligne–Mumford stack with separated diagonal and finite stabilizers.*

In either case the same conclusions hold for the restriction $\eta|_{\mathcal{Z}}$ of η to any open and closed substack $\mathcal{Z} \subseteq \mathcal{Y}^T$.

Proof. *Proof of (a).* Since $\mathcal{Y}$ is locally of finite type with affine stabilizers, $\mathcal{Y}^{hT}$ is a derived 1-Artin stack with affine stabilizers [Ara+25, Thm. A.20] locally almost of finite presentation over k [Ara+25, Thm. 5.1.1.]. By Remark 2.6, $\mathcal{Y}^T$ is therefore an algebraic stack of locally finite presentation over k with affine stabilizers. Every morphism in $\mathcal{Y}^T$ is a morphism in $\mathcal{Y}$ (Remark 2.2), so η is faithful and therefore representable by algebraic spaces by [Stacks, Tag 04Y5]. Morphisms representable by algebraic spaces are DM-type [Stacks, Tag 050E].

It remains to show that η is separated. Let $y : S \rightarrow \mathcal{Y}$ be a point of $\mathcal{Y}(S)$ with S a scheme and set $Y = S \times_{\mathcal{Y}} \mathcal{Y}^T$. Let W be an S -scheme and $s_1, s_2 \in Y(W)$ sections of $\text{St}_y \times_S W \rightarrow T_W$, group-theoretic sections by [Rom22, Lemma. 4.1.2]. The morphism $\text{St}_y \times_S W \rightarrow T_W$ is a base change of $\Delta_{\mathcal{Y}}$ and hence separated, so s_1 and s_2 agree exactly on a closed subscheme $E \subseteq T_W$. The equalizer of s_1 and s_2 in W is the base change of $\Delta_{Y/S}$ along $(s_1, s_2) : W \rightarrow Y \times_S Y$, the Weil restriction $\prod_{T_W/W} E$. This is a closed subscheme of W by [Rom22, Thm. A.6], hence $\Delta_{Y/S}$ is a closed immersion and η is separated. Compare this to [Rom22, Thm. 4.1.4].

Proof of (b). By [Ara+25, Prop. A.23], applied to $\mathcal{Y}$ regarded as a dervied stack, $\varepsilon : \mathcal{Y}^{hT} \rightarrow \mathcal{Y}$ is a closed immersion. Classical truncation preserves closed immersions, so η is a closed immersion by Remark 2.6 and $\mathcal{Y}^T$ is Deligne-Mumford. The diagonal $\Delta_{\mathcal{Y}^T/\mathcal{Y}}$ is an isomorphism since η is an isomorphism, and because $\Delta_{\mathcal{Y}^T}$ is the base change of $\Delta_{\mathcal{Y}^T/\mathcal{Y}}$ along $\eta \times \eta$, it is quasi-compact and separated.

In either case, if $\mathcal{Z} \subseteq \mathcal{Y}^T$ is open and closed then $\eta|_{\mathcal{Z}}$ is the composite of the open and closed immersion $\mathcal{Z} \hookrightarrow \mathcal{Y}^T$ with η , and each of the properties above is preserved by composing with an open and closed immersion. $\square$

Two further facts let us pass the hypotheses of Proposition 2.7 to the stacks built from $\mathcal{Y}$ in Sections 3 and 4: fixed stacks commute with fiber products, and the hypotheses of part (a) descend along representable separated morphisms.

Lemma 2.8. *Let $A_1 \rightarrow A_0 \leftarrow A_2$ be T -equivariant morphisms of T -stacks. Then $A_1 \times_{A_0} A_2$ has a T -action for which the projections are equivariant, and $(A_1 \times_{A_0} A_2)^T \simeq A_1^T \times_{A_0^T} A_2^T$.*

Proof. By [Rom05, Def. 2.3] the fixed stack $\mathcal{M}^G$ 2-represents $\mathcal{H}om_{G\text{-st}}(\iota(-), \mathcal{M})$, and the 2-category of G -stacks has fibred products, for which the projections are G -equivariant (see the paragraph preceding [Rom05, Def. 2.3]). Since $\mathcal{H}om_{G\text{-st}}(\iota(N), -)$ preserves fibred products, $(A_1 \times_{A_0} A_2)^T$ and $A_1^T \times_{A_0^T} A_2^T$ 2-represent the same 2-functor. $\square$

Lemma 2.9 (Transfer Lemma). *Let $f : \mathcal{Y}' \rightarrow \mathcal{Y}$ be a morphism of algebraic stacks over k , representable by algebraic spaces and separated. If $\mathcal{Y}$ has quasi-compact separated diagonal and affine stabilizers, then so does $\mathcal{Y}'$.*

Proof. The relative diagonal $\Delta_{\mathcal{Y}'/\mathcal{Y}}$ is a closed immersion since f is representable and separated and $\Delta_{\mathcal{Y}'} = \rho \circ \Delta_{\mathcal{Y}'/\mathcal{Y}}$ with ρ the base change of $\Delta_{\mathcal{Y}}$ along $f \times f$. A closed immersion followed by a base change of a quasi-compact separated morphism is quasi-compact and separated. Pulling the factorization back along a geometric point (y', y') of $\mathcal{Y}' \times \mathcal{Y}'$ exhibits $\text{Aut}(y') \rightarrow \text{Aut}(f(y'))$ as a base change of $\Delta_{\mathcal{Y}'/\mathcal{Y}}$, hence a closed immersion. A closed subgroup of an affine group is also affine. $\square$

2.3. Deformation theory of the fixed stack. We now resolve one of the problems highlighted in the introduction: given a T -equivariant morphism $M \rightarrow \mathcal{Y}$ from a Deligne-Mumford stack to an algebraic stack equipped with a T -equivariant perfect obstruction theory $E \rightarrow L_{M/\mathcal{Y}}$, construct a perfect obstruction theory for $M^T \rightarrow \mathcal{Y}^T$ compatible with E . The situation turns out to be perfectly analogous to the one in [GP99]; the fixed part of $L_{\mathcal{Y}^T/\mathcal{Y}}$ vanishes in degrees -1 and 0 (Lemma 2.11) and $(E|_{M^T})^{\text{fix}}$ is a perfect obstruction theory for $\pi^T : M^T \rightarrow \mathcal{Y}^T$ (Lemma 2.12).

Remark 2.10. Let $\mathcal{Y}$ be an algebraic stack over k with a T -action for which $\mathcal{Y}^T$ is algebraic. The T -equivariant structure on $L_{\mathcal{Y}^T/\mathcal{Y}}$ is, by definition, $L_{\eta_T} \in D_{qc}(\mathcal{Y}^T \times BT)$ where $\eta_T : \mathcal{Y}^T \times BT \rightarrow [\mathcal{Y}/T]$ is the morphism over BT induced by η and its universal fixed point structure. Writing $e : \mathcal{Y}^{hT} \times BT \rightarrow [\mathcal{Y}/T]$ for the quotient of $\varepsilon : \mathcal{Y}^{hT} \rightarrow \mathcal{Y}$ by T as in [Ara+25, Cor. A.27], we get $\eta_T = e \circ (t \times \text{id}_{BT})$ by Remark 2.6, and hence this equivariant structure is the restriction along $t \times \text{id}_{BT}$ of the one on $L_{\mathcal{Y}^{hT}/\mathcal{Y}}$. The structure matters: the pullback of L_η along the projection $\mathcal{Y}^T \times BT \rightarrow \mathcal{Y}^T$ results in an another T -equivariant structure entirely of weight zero on $\mathcal{Y}^T$, for which the conclusion of Lemma 2.11 is obviously false.

Lemma 2.11. *If T is a diagonalizable group over a field which acts on an algebraic stack locally of finite type over k with affine stabilizers and quasi-compact separated diagonal, then the two term truncation of $L_{\mathcal{Y}^T/\mathcal{Y}}$ has no fixed part with respect to the natural T -action.*

Proof. We work directly with the complexes on $\mathcal{Y}^T \times BT$. The deformation theory of $\mathcal{Y}^{hT}$ is computed by [Ara+25, Cor. A.27, A.30, A.31] as the complex of T -invariants of $\varepsilon^* L_{\mathcal{Y}}$, and the relevant consequences for us are

$$(8) \quad L_{\mathcal{Y}^{hT}} \simeq (\varepsilon^* L_{\mathcal{Y}})^{\text{fix}}, \quad L_e \simeq (\varepsilon^* L_{\mathcal{Y}})^{\text{mov}}[1] \quad \text{in } D_{qc}(\mathcal{Y}^{hT} \times BT),$$

with the notation from Remark 2.10. In particular, L_e has no weight-zero part. The exact triangle resulting from the factorization of η_T is

$$(t \times \text{id}_{BT})^* L_e \rightarrow L_{\eta_T} \rightarrow L_{t \times \text{id}_{BT}} \xrightarrow{+1} \dots,$$

and taking the fixed part of the long exact sequence in cohomology yields

$$0 \rightarrow h^{-1}(L_{\eta_T})^{\text{fix}} \rightarrow h^{-1}(L_{t \times \text{id}_{BT}})^{\text{fix}} \rightarrow 0 \rightarrow h^0(L_{\eta_T})^{\text{fix}} \rightarrow h^0(L_{t \times \text{id}_{BT}})^{\text{fix}} \rightarrow 0.$$

and gives isomorphisms therefore gives isomorphisms $h^i(L_{\eta_T}) \cong h^i(L_{t \times \text{id}_{BT}})$ for $i = -1, 0$.

We conclude using a fact about truncations in derived algebraic geometry: truncation changes the cotangent complex only in degrees ≤ -2 [STV15, Prop. 1.2], that is, for any derived Artin stack $\mathcal{X}$, $L_{\mathcal{X}_{\text{cl}}/\mathcal{X}} \in D^{\leq -2}(\mathcal{X}_{\text{cl}})$. Since $t \times \text{id}_{BT}$ is precisely the truncation of $\mathcal{Y}^{hT} \times BT$ to the classical $\mathcal{Y}^T \times BT$, $h^i(L_{t \times \text{id}_{BT}})$ vanishes for $i = -1, 0$, and we are done. $\square$

We end this section describing how, given an obstruction theory for $\pi : M \rightarrow \mathcal{Y}$, we obtain a canonical obstruction theory for $M^T \rightarrow \mathcal{Y}^T$.

Lemma 2.12. *Suppose E is a T -equivariant relative perfect obstruction theory for $\pi : M \rightarrow \mathcal{Y}$ where M is a Deligne-Mumford stack and $\mathcal{Y}$ is an algebraic stack. That is, E is an element in the T -equivariant derived category of M of perfect amplitude $[-1, 0]$ together with a T -equivariant morphism $\phi : E \rightarrow L_{M/\mathcal{Y}}$ which is an*

obstruction theory in the sense of [BF97, Definition 4.4]. If $\mathcal{Y}^T$ is an algebraic stack, then $(E|_{M^T})^f$ is a canonical relative perfect obstruction theory for M^T over $\mathcal{Y}^T$.

Proof. We have a commutative square (not Cartesian)

$$\begin{array}{ccc} M^T & \xrightarrow{\iota} & M \\ \downarrow \pi^T & & \downarrow \pi \\ \mathcal{Y}^T & \xrightarrow{\eta} & \mathcal{Y} \end{array}.$$

Traversing from M^T to $\mathcal{Y}$ through M and through $\mathcal{Y}^T$ each gives us a transitivity triangle of cotangent complexes, and pulling back ϕ by ι^* gives us a morphism $\iota^*\phi : \iota^*E \rightarrow \iota^*L_{M/\mathcal{Y}}$. We can arrange this data as follows:

$$\begin{array}{ccccccc} \iota^*E & & & & & & \\ \downarrow \iota^*\phi & & & & & & \\ \iota^*L_{M/\mathcal{Y}} & \xrightarrow{\alpha} & L_{M^T/\mathcal{Y}} & \longrightarrow & L_{M^T/M} & \xrightarrow{+1} & \dots \\ & & \parallel & & & & \\ (\pi^T)^*L_{\mathcal{Y}^T/\mathcal{Y}} & \longrightarrow & L_{M^T/\mathcal{Y}} & \xrightarrow{\beta} & L_{M^T/\mathcal{Y}^T} & \xrightarrow{+1} & \dots \end{array}$$

where the middle equality follows from the commutativity of the diagram. Write ι^*E as $E|_{M^T}$ to match the notation in the statement and define ψ to be the composition of $\iota^*\phi$, α and β after taking fixed parts:

$$\psi : (E|_{M^T})^f \xrightarrow{\iota^*\phi^f} (\iota^*L_{M/\mathcal{Y}})^f \xrightarrow{\alpha^f} (L_{M^T/\mathcal{Y}})^f \xrightarrow{\beta^f} (L_{M^T/\mathcal{Y}^T})^f.$$

We show that ψ is a perfect obstruction theory. First consider the map $\iota^*\phi$. Right exactness of derived pullback implies that $h^0(\iota^*E) = \iota^*h^0(E)$ and $h^0(\iota^*L_{M/\mathcal{Y}}) = \iota^*h^0(L_{M/\mathcal{Y}})$, and ι^* preserves isomorphisms, so $h^0(\iota^*\phi)$ is an isomorphism. For $h^{-1}(-)$, we get a Tor correction and a diagram with exact rows:

$$\begin{array}{ccccccc} \iota^*h^{-1}(E) & \longrightarrow & h^{-1}(\iota^*E) & \longrightarrow & \mathrm{Tor}_1(h^0(E)) & \longrightarrow & 0 \\ \downarrow \iota^*h^{-1}(\phi) & & \downarrow h^{-1}(\iota^*\phi) & & \downarrow \mathrm{Tor}_1(h^0(\phi)) & & \downarrow \\ \iota^*h^{-1}(L_{M/\mathcal{Y}}) & \longrightarrow & h^{-1}(\iota^*L_{M/\mathcal{Y}}) & \longrightarrow & \mathrm{Tor}_1(h^0(L_{M/\mathcal{Y}})) & \longrightarrow & 0 \end{array}$$

Tor preserves isomorphisms so $\mathrm{Tor}_1(h^0(\phi))$ is an isomorphism. The morphism $h^{-1}(\phi)$ is surjective by assumption and $\iota^*(-)$ is right exact, so $\iota^*h^{-1}(\phi)$ is also surjective. Applying [Stacks, Lemma 05QA] then gives us that $h^{-1}\iota^*\phi$ is surjective as well.

Now consider the map α . The fixed portion of the long exact sequence of the transitivity triangle involving α reads

$$\begin{aligned} h^{-1}(\iota^* L_{M/\mathcal{Y}})^f &\xrightarrow{h^{-1}(\alpha^f)} h^{-1}(L_{M^T/\mathcal{Y}})^f \rightarrow h^{-1}(L_{M^T/M})^f \rightarrow \dots \\ \dots \rightarrow h^0(\iota^* L_{M/\mathcal{Y}})^f &\xrightarrow{h^0(\alpha^f)} h^0(L_{M^T/\mathcal{Y}})^f \rightarrow h^0(L_{M^T/M})^f \rightarrow 0 \end{aligned}$$

Since ι is a closed immersion (Proposition 2.7(b)) $h^0(L_{M^T/M}) = \Omega_{M^T/M} = 0$, and applying Lemma 2.11 to $\mathcal{M}$ gives us $h^{-1}(L_{M^T/M})^f = 0$. This implies $h^0(\alpha^f)$ is an isomorphism and $h^{-1}(\alpha^f)$ is surjective.

Finally consider β . Because T acts trivially on $\mathcal{M}^T$ and $\mathcal{Y}^T$, $L_{\mathcal{M}^T/\mathcal{Y}^T}$ is of pure weight zero and the weight-zero part of the long exact sequence of the bottom triangle reads

$$\begin{aligned} h^{-1}(\iota^* L_{\mathcal{Y}^T/\mathcal{Y}})^f &\rightarrow h^{-1}(L_{M^T/\mathcal{Y}})^f \xrightarrow{h^{-1}(\beta^f)} h^{-1}(L_{M^T/\mathcal{Y}^T})^f \rightarrow \dots \\ \dots \rightarrow h^0(\iota^* L_{\mathcal{Y}^T/\mathcal{Y}})^f &\rightarrow h^0(L_{M^T/\mathcal{Y}})^f \xrightarrow{h^0(\beta^f)} h^0(L_{M^T/\mathcal{Y}^T})^f \rightarrow 0. \end{aligned}$$

Since $(\pi^T)^*$ is right t -exact and compatible with weights,

$$h^0((\pi^T)^* L_{\mathcal{Y}^T/\mathcal{Y}})^f = (\pi^T)^*(\Omega_{\mathcal{Y}^T/\mathcal{Y}}^f) = 0$$

by Lemma 2.11. Hence $h^0(\beta^f)$ is an isomorphism and $h^{-1}(\beta^f)$ is surjective. Thus $h^0(\psi)$ is the composite of isomorphisms and $h^{-1}(\psi)$ is the composite of surjections, so we conclude that ψ is a perfect obstruction theory. $\square$

2.4. The fixed stack need not be algebraic. Proposition 2.7 is stated over a field, and over a field affine stabilizers suffice for the algebraicity of $\mathcal{Y}^T$ [Ara+25, Theorem A.20]. However, over a base which is *not* a field the affine-diagonal hypothesis of Romagny's theorem [Rom22, Theorem 1.2.1] cannot simply be dropped. Example 2.13 is a reconstruction of an example mentioned by Romagny following [Rom22, Theorem 1.2.2] and found in [SGA3-II, Exp. IX, Rem. 7.4]: over a discrete valuation ring, $\mathcal{Y}^T$ may fail to be algebraic when $\mathcal{Y}$ does not have affine diagonal. The stack in the example does not have affine stabilizers either, so it says nothing about whether the hypotheses of Proposition 2.7(a) can be weakened; we do not know what the minimal hypotheses are.

Example 2.13. Let $R = \mathbb{C}[[q]]$, $K = \mathbb{C}((q))$, $S = \text{Spec } R$ and let $W \rightarrow S$ be the Weierstrass model

$$y^2 + xy = x^3 + a_4x + a_6$$

of the Tate curve [Sil99]. Now let E be the smooth locus of W . Then E is a quasi-compact open subscheme of the projective S -scheme W , hence $E \rightarrow S$ is separated, finitely presented and smooth. Its generic fiber is an elliptic curve and its central fiber is the smooth locus of a nodal cubic, so $E_{\mathbb{C}} = \mathbb{G}_m$.

Set $\mathcal{Y} = BE = [S/E]$ and let $\mathbb{G}_m$ act trivially on $\mathcal{Y}$. Then of the hypotheses in [Rom22, Theorem 1.2.1], $\mathcal{Y}$ only fails the affine diagonal requirement. Indeed, $\mathcal{Y}$ is an algebraic stack by [Stacks, 06PL] and the group $\mathbb{G}_m$ is reductive. However, if we base change $\Delta_{BE} : BE \rightarrow BE \times_S BE$ along $S \rightarrow BE \times_S BE$ given in each component by the trivial torsor $S \rightarrow BE$, then we get

$$BE \times_{BE \times_S BE} S \cong E.$$

The fiber of E over the generic point of S is the proper curve $E_K \rightarrow \text{Spec } K$, so E_K is not affine and neither is Δ_{BE} .

We claim that $(BE)^{\mathbb{G}_m}$ is not algebraic. The Romagny fixed stack is

$$\mathcal{Y}^{\mathbb{G}_m} \cong BE \times_S F, \quad F = \underline{\text{Hom}}_{S\text{-gp}}(\mathbb{G}_m, E),$$

and $\eta : \mathcal{Y}^{\mathbb{G}_m} \rightarrow \mathcal{Y}$ is the projection to the first factor by the description at the end of Remark 2.2. If $\mathcal{Y}^{\mathbb{G}_m}$ were algebraic, then η , being faithful (see the proof of Proposition 2.7(a)), would be representable by algebraic spaces [Stacks, Tag 04Y5], so its base change $F = S \times_{BE} \mathcal{Y}^{\mathbb{G}_m}$ along the trivial torsor would be an algebraic space over S . We show that no algebraic space can have the values F has by exhibiting a compatible family $f_n \in F(A_n)$ for $A_n \in R/(q^n)$ with $f_1 \neq 0$, while $F(R) = 0$.

This is straightforward. Observe first that any group homomorphism $f : \mathbb{G}_{m,R} \rightarrow E$ must induce a group homomorphism $f_K : \mathbb{G}_{m,K} \rightarrow E_K$. This extends to a morphism $\mathbb{P}_K^1 \rightarrow E_K$, but the only such morphism is constant, hence f_K is trivial and $F(R) = 0$. If we instead reduce to $\mathbb{C} = R/(q)$, then we get

$$F(\mathbb{C}) = \text{Hom}(\mathbb{G}_m, \mathbb{G}_m) = \mathbb{Z}.$$

Next we need to show there exists some non-trivial element in $\lim \text{Hom}(\mathbb{G}_{m,A_n}, E_{A_n})$ for $A_n = R/(q^n)$. One can do this directly by showing the identity morphism $f_1 : \mathbb{G}_{m,\mathbb{C}} \rightarrow \mathbb{G}_{m,\mathbb{C}} = E_{\mathbb{C}}$ lifts to a morphism $f_n : \mathbb{G}_{m,A_n} \rightarrow E_{A_n}$ where $A_n = R/(q^n)$. In homogeneous coordinates, it's given by

$$f_n(t) = [t(1-t) + (1-t)^3 P_n(t, q) : t^2 + (1-t)^3 Q_n(t, q) : (1-t)^3]$$

for certain Laurent polynomials $P_n, Q_n \in qA_n[t, t^{-1}]$ obtained by truncating the Tate series modulo q^n .

Suppose now that $\mathcal{Y}^{\mathbb{G}_m}$ is algebraic, so that F is an algebraic space over S , and choose an étale surjection $p : V \rightarrow F$ from a scheme. The $\mathbb{C}$ -point $f_1 \in F(A_1)$ lifts to a $\mathbb{C}$ -point v of V , since p is surjective and $\mathbb{C}$ is algebraically closed. As p is étale, hence formally étale, and $\text{Spec } A_{n-1} \subset \text{Spec } A_n$ is a nilpotent thickening, the maps $f_n : \text{Spec } A_n \rightarrow F$ lift uniquely and compatibly to $\lambda_n : \text{Spec } A_n \rightarrow V$ with $\lambda_1 = v$. A local Artinian scheme lands in every open neighbourhood of its closed point, so all λ_n factor through a single affine open $\text{Spec } B \ni v$ of V ; they are then given by compatible ring maps $B \rightarrow A_n$, that is, by a ring map $B \rightarrow \varprojlim A_n = R$. The

resulting composite $\text{Spec } R \rightarrow V \rightarrow F$ is an element $f \in F(R)$ with $f|_{A_n} = f_n$ for every n . But $F(R) = 0$, so $f_1 = 0$, contradicting $f_1 = \text{id}_{G_m}$. Hence $\mathcal{Y}^{G_m} = (BE)^{G_m}$ is not algebraic.

3. THE RESIDUE MAP FOR ARTIN STACKS

Consider the absolute situation for a moment, where M is a Deligne-Mumford stack over a field k , $T = (G_{m,k}^n)$ is an algebraic torus acting on M and trivially on $\text{Spec } k$, and E is a T -equivariant perfect obstruction theory for the map $\pi : M \rightarrow \text{Spec } k$. From this data one can write down a perfect obstruction theory E^f for M^T over $\text{Spec } k$, allowing one to form both $[M]^{vir}$ and $[M^T]^{vir}$ via virtual pullback over π and π^T respectively. The content of virtual localization is then in studying the pushforward of $\iota : M^T \rightarrow M$ in (localized) equivariant Chow groups together with its inverse $\text{Res} : A_*^T(M) \otimes Q_T \rightarrow A_*^T(M^T) \otimes Q_T$.

Edidin and Graham prove two things in this situation [EG98]. First, for any scheme X of finite type with a T -action, the pushforward $i_* : A_*^T(X^T) \rightarrow A_*^T(X)$ becomes an isomorphism after inverting the nonzero characters of T [EG98, Thm. 1]. We write Q_T for this localization of $R_T = A_T^*(\text{pt})$ and $\text{Res}_{X^T/X}$ for the inverse of i_* . Second, they make the inverse explicit. When X is smooth, the normal bundle $N_{F/X}$ of a component F of X^T is moving, its Euler class is a unit in $A_T^*(F)_{Q_T}$ and the F -component of $\text{Res}_{X^T/X}$ is $e_T(N_{F/X})^{-1} \cap i_F^*$ [EG98, Thm. 2]. When X is singular the normal cone is not a bundle, but if X embeds equivariantly in a smooth T -variety the same formula holds with the normal bundle of F in the ambient variety [EG98, Prop. 6]. The common mechanism is the following: embed the normal cone of $Z \subseteq X^T$ in a moving vector bundle V , specialize to the cone, push into V , and pull back along the zero section; the result is $e_T(V) \cap \text{Res}_{Z/X}$.

Proposition 3.1 ([EG98]). *Suppose X is a scheme of finite type over k with a T -action and let $\iota : Z \hookrightarrow X$ be the inclusion of an open and closed subscheme of X^T . Write $\text{Res}_{Z/X} : A_*^T(X)_{Q_T} \rightarrow A_*^T(Z)_{Q_T}$ for the Z -component of the inverse of the isomorphism $i_* : A_*^T(X^T)_{Q^T} \rightarrow A_*^T(X)_{Q^T}$. Suppose the normal cone $C_{X/Z}$ admits a T -equivariant embedding $j : C \rightarrow V$ into a moving vector bundle over Z , then we have two maps $A_*^T(X) \rightarrow A_*^T(Z)_Q$:*

$$\begin{array}{c} \text{Res}_{Z/X} \\ \curvearrowright \\ A_*^T(X)_Q \xrightarrow{\sigma} A_*^T(C)_Q \xrightarrow{j_*} A_*^T(V)_Q \xrightarrow{s_V^*} A_*^T(Z)_Q \end{array}$$

The composite of the specialization map σ , the pushforward j_* and the Gysin pullback s_V^* of the zero section satisfy the following identity with $\text{Res}_{Z/X}$:

$$(9) \quad e(V) \cap \text{Res}_{Z/X}(\alpha) = (s_V^* \circ j_* \circ \sigma)(\alpha).$$

Now consider the relative situation. We start with a T -equivariant map $\pi : M \rightarrow \mathcal{Y}$ from a Deligne-Mumford stack M to an Artin stack $\mathcal{Y}$ together with a relative perfect obstruction theory E for π . We can still form a corresponding canonical perfect obstruction theory E^f for $\pi^T : M^T \rightarrow \mathcal{Y}^T$ by Lemma 2.12, but crucially, the bases of these two obstruction theories are different. This means that before studying $\iota : M^T \rightarrow M$, we must relate classes in $\mathcal{Y}$ to classes in $\mathcal{Y}^T$. At a glance this is exactly the role of the residue map, but there is a problem: the main localization theorem fails spectacularly for Artin stacks. Consider Example 2.5 for instance. If $\mathcal{Y} = B\mathbb{G}_m$ with the trivial $T = \mathbb{G}_m$ action, then $\mathcal{Y}^T = \coprod_{\chi \in \mathbb{Z}} B\mathbb{G}_m$ and $A_*^T(\mathcal{Y}^T) \cong \bigoplus_{\chi \in \mathbb{Z}} A_*^T(\mathcal{Y})$, hence i_* fails to become an isomorphism after *any* localization.

Rearranging the identity 9 from Proposition 3.1 gives us

$$(10) \quad \text{Res}_{Z/X}(\alpha) = e_T(V)^{-1} \cap (s_V^* \circ j_* \circ \sigma)(\alpha).$$

Supposing we can make sense of all terms on the right hand side, we can take this as the definition for Res . The scheme X becomes an Artin stack $\mathcal{Y}$, Z becomes the fixed stack $\mathcal{Y}^T$, the cone C is replaced by the intrinsic normal cone stack $\mathfrak{C}_{\mathcal{Y}^T/\mathcal{Y}}$ of η ([Man12, Definition 2.30]) and V becomes a vector bundle stack $\mathfrak{E}$ over $\mathcal{Y}^T$. We then have two problems, and they are the subject of this section.

Problem 1: (The domain of Res .) The fixed stack $\mathcal{Y}^T$ is rarely of finite type over k , even when $\mathcal{Y}$ is (Example 2.5), and Kresch's Chow groups are only defined for stacks of finite type with a stratification by global quotients. We therefore define the residue not on $\mathcal{Y}^T$ but on an open and closed substack $\mathcal{Z} \subseteq \mathcal{Y}^T$ of finite type; in the application $\mathcal{Z}$ is any such substack containing the image of M^T , and the localization formula of Section 4 does not see which one is chosen. The hypotheses of Convention 1.1 (affine stabilizers, quasi-compact separated diagonal) are there to guarantee the stratification, and hence that the Gysin map of a vector bundle stack over $\mathcal{Z}$ exists [Kre99, Prop. 5.3.2].

Problem 2: (The Euler class $e_T^{-1}(\mathfrak{E})$ and its inverse.) Suppose we have a finite-type scheme X fixed pointwise by the action of $T = \mathbb{G}_m$, so that $A_T^*(\text{pt}) = \mathbb{Q}[t]$ and, the action being trivial, $A_T^*(X) = A_T^*(\text{pt}) \otimes A^*(X)$ by the projective bundle formula applied to $X \times \mathbb{P}^N$. The equivariant Euler class of a moving line bundle L on X is then

$$e_T(L) = c_1^T(L) = wt + c$$

where $w \in \mathbb{Z} \setminus \{0\}$ is the weight of the T -action on L and $c = c_1(L) \in A^1(X)$ is the ordinary first Chern class of L acting on $A_*(X)$. The formal inverse is then

$$e_T^{-1}(L) = (wt)^{-1} - c(wt)^{-2} + c^2(wt)^{-3} - \dots$$

When X is a scheme, $A_n(X) = 0$ for $n < 0$. This means c acts nilpotently on $A_*(X)$, so this sum is actually finite. To invert $e_T(L)$ it therefore suffices to invert t , which is why $e_T^{-1}(L)$ acts on $A_*^T(X)_{Q_T}$.

However if $\mathcal{Y}$ is instead an algebraic stack it is no longer the case that c acts nilpotently; $A_n(\mathcal{Y})$ can be nonzero for arbitrarily negative n . When T is rank 1, we can solve this by working in the ring of formal Laurent series in t^{-1} , namely,

$$A_T^*(\mathcal{Y})_{\text{loc}} := A_T^*(\mathcal{Y})_{Q_T}[[t^{-1}]].$$

It is more complicated to give a neat presentation of these groups for higher rank tori. For our purposes it will suffice to take $A_T^*(\mathcal{Y})_{\text{loc}}$ to be $A_T^*(\mathcal{Y})_{Q_T}$ with “all necessary Euler classes inverted”. We will give a more precise description in the following section (§ 3.1).

Problem 3: (Dependence on the embedding $\mathfrak{E} \hookrightarrow \mathfrak{E}$.) If η is DM-type and we fix an embedding $j : \mathfrak{E}_{\mathcal{Y}^T/\mathcal{Y}} \hookrightarrow \mathfrak{E}$, then the composition $s_{\mathfrak{E}}^* \circ j_* \circ \sigma$ is simply Manolache’s virtual pullback $\eta^!$. The existence of data $j : \mathfrak{E}_{\mathcal{Y}^T/\mathcal{Y}} \hookrightarrow \mathfrak{E}$ is equivalent to the existence of a perfect relative obstruction theory $E_{\mathcal{Y}^T/\mathcal{Y}}^\bullet \rightarrow L_{\mathcal{Y}^T/\mathcal{Y}}$, in which case $\mathfrak{E} = h^1/h^0(E_{\mathcal{Y}^T/\mathcal{Y}}^\bullet)$ [Man12, Proposition 3.11]. There is no canonical choice of perfect obstruction theory for η , which is a problem since $\eta^!$ depends on this choice. We solve this problem in § 3.2.

3.1. Euler classes of vector bundle stacks. In this section we denote vector bundle stacks with ordinary letters V rather than fraktur letters for readability.

Convention (the localized group.) Let T be a split diagonalizable group, $R_T = A_T^*(\text{pt})$. For an algebraic stack $\mathcal{Y}$ with a trivial T action, let S denote the multiplicative set of operators on $A_*^T(\mathcal{Y})$ generated by the equivariant Euler classes $e_T(V)$ of finite rank moving T -equivariant vector bundles V on $\mathcal{Y}$. We then define

$$A_T^*(\mathcal{Y})_{\text{loc}} = S^{-1}A_*^T(\mathcal{Y}).$$

For $T = \mathbb{G}_m$ this localization maps to the ring

$$A_*(\mathcal{Y})((t^{-1})) = A_*(\mathcal{Y})[[t]][t]$$

of formal Laurent series in t^{-1} by expanding

$$e_T(L)^{-1} = (wt + c)^{-1} = \sum_{k \geq 0} (-1)^k c^k (wt)^{-k-1}.$$

Definition 3.2. Let $\mathcal{Y}$ be an algebraic stack with a trivial T -action and let V be a vector bundle stack over $\mathcal{Y}$. We say that V is *finite rank and moving* if, up to equivalence of vector bundle stacks, it can be represented as a quotient $[V_1/V_0]$ where

V_0 and V_1 are each finite rank moving vector bundles over $\mathcal{Y}$. If V is such a vector bundle stack then we then define its Euler class to be

$$e_T(V) = e_T(V_1)/e_T(V_0) \in A_T^*(\mathcal{Y})_{\text{loc}}.$$

This definition is meaningless without the following lemma.

Lemma 3.3. *Let T be a split diagonalizable group and let $\mathcal{Y}$ be an algebraic stack with trivial T -action. Let*

$$E^\bullet = [E^0 \rightarrow E^1], \quad F^\bullet = [F^0 \rightarrow F^1]$$

be a two-term complexes of finite-rank moving T -equivariant locally free sheaves. If $E^\bullet$ and $F^\bullet$ are isomorphic in the equivariant derived category, then

$$e_T(E^1)e_T(F^0) = e_T(E^0)e_T(F^1)$$

in $A_T^(\mathcal{Y})$. In particular, $e_T(E^\bullet) := e_T(E^1)e_T(E_0)^{-1}$ and $e_T(F^\bullet)$ agree in $A_*^T(\mathcal{Y})_{\text{loc}}$*

Proof. Fix a complex $G_\bullet = [G_0 \xrightarrow{d} G_1]$ of quasicoherent sheaves and morphisms $\phi : E_\bullet \rightarrow G_\bullet$ and $\psi : F_\bullet \rightarrow G_\bullet$ exhibiting the isomorphism. Given the data of a locally free sheaf A_1 and a morphism $\chi : A_1 \rightarrow G_1$ such that the induced map $\bar{\chi} : A_1 \rightarrow H^1(G_\bullet)$ is surjective, we can find a two-term complex $A_\bullet^\chi$ of locally free sheaves which is quasi-isomorphic to $G_\bullet$. We do this because while we can't take the Euler class of $G_\bullet$, we can take the Euler class of $A_\bullet^\chi$.

First, consider the map $\chi - d : A_1 \oplus G_0 \rightarrow G_1$ and set $A_0 = \ker(\chi - d)$. Then define $A_\bullet^\chi = [A_0 \xrightarrow{d_A} A_1]$, where d_A is the composition $A_0 \hookrightarrow A_1 \oplus G_0 \xrightarrow{\text{proj}_1} A_1$. We have an obvious map of complexes $\tilde{\chi} : A_\bullet^\chi \rightarrow G_\bullet$ whose mapping cone $0 \rightarrow A_0 \rightarrow A_1 \oplus G_0 \rightarrow G_1 \rightarrow 0$ is exact because χ induces a surjection on the first cohomology of $G_\bullet$, hence $A_\bullet^\chi$ is quasi-isomorphic to $G_\bullet$. The complex $G_\bullet$ is perfect by definition — it is quasi-isomorphic to $E_\bullet$, a bounded complex of locally free sheaves — so $A_\bullet^\chi$ is perfect as well. Whenever the second term of a perfect two-term complex is locally free, so is its first term, and therefore $A_\bullet^\chi$ consists of locally free sheaves as desired.

The sheaf $E_1 \oplus F_1$ admits two obvious maps to G_1 inducing surjections onto the first cohomology of G_1 . They are $\phi \oplus 0$ and $0 \oplus \psi$, and using the process above, we obtain identities $e(E_\bullet) = e(A_\bullet^{\phi \oplus 0})$ and $e(F_\bullet) = e(A_\bullet^{0 \oplus \psi})$ respectively. The locus of maps in the vector space $\text{Hom}(E_1 \oplus F_1, G_1)$ inducing a surjection in cohomology is open in the Zariski topology, therefore, we can find a family of Chow classes parameterized by some $U \subset \text{Hom}(E_1 \oplus F_1, G_1)$ containing $e(E_\bullet)$ and $e(F_\bullet)$. This gives us our result. $\square$

3.2. The residue map. We now define a version of the residue map which will serve as the bridge between invariants of $\mathcal{Y}$ and invariants of $\mathcal{Y}^T$. We'll prove it for an open and closed substack $\mathcal{Z} \subseteq \mathcal{Y}^T$, open so that we may interchange $\tau_{\geq -1} L_\eta|_{\mathcal{Z}}$

with $\tau_{\geq -1}L_{\eta|_{\mathcal{Z}}}$ and closed so that the inclusion $\mathcal{Z} \hookrightarrow \mathcal{Y}^T$ is a closed immersion. We abuse notation and write η for both $\eta : \mathcal{Y}^T \rightarrow \mathcal{Y}$ and $\eta|_{\mathcal{Z}} : \mathcal{Z} \rightarrow \mathcal{Y}$.

Taking inspiration from the Edidin-Graham case in Proposition 3.1, the map $\text{Res}_{\mathcal{Z}/\mathcal{Y}} : A_*^T(\mathcal{Y}) \rightarrow A_*^T(\mathcal{Z})_{\text{loc}}$ we wish to write down should be

$$(11) \quad \text{Res}_{\mathcal{Z}/\mathcal{Y}}(\alpha) = e_T^{-1}(\mathfrak{E}) \cap \eta_{\mathfrak{E}}^!(\alpha)$$

where $\eta_{\mathfrak{E}}^!$ is virtual pullback. Virtual pullback requires the specification of additional data, namely, $(j, \mathfrak{E})$ where $\mathfrak{E}$ is a vector bundle stack over $\mathcal{Z}$ and $j : \mathfrak{C}_{\mathcal{Z}/\mathcal{Y}} \hookrightarrow \mathfrak{E}$ is a closed embedding of the intrinsic normal cone of $\eta : \mathcal{Z} \rightarrow \mathcal{Y}$. For this to make sense equivariantly we additionally need that $\mathfrak{E}$ is T -equivariant, and to ensure $e_T(\mathfrak{E})$ is a unit we require that $\mathfrak{E}$ is moving. We'll refer to $(j, \mathfrak{E})$ as *embedding data*.

Given a choice of embedding data $(j, \mathfrak{E})$ for the cone stack $\mathfrak{C}_{\eta}$, the equivariant virtual pullback $\eta^!$ [Man12] is the composite

$$\eta^! : A_*^T(\mathcal{Y}) \xrightarrow{\sigma} A_*^T(\mathfrak{C}_{\eta}) \xrightarrow{j_*} A_*^T(\mathfrak{E}) \xrightarrow{0_{\mathfrak{E}}^!} A_*^T(\mathcal{Z})$$

where σ is the equivariant specialization map on $\mathcal{Z} \rightarrow \mathcal{Y}$, j_* is equivariant projective pushforward, $0_{\mathfrak{E}}^!$ is the Gysin map for vector bundle stacks. We make a comment about this last map.

Convention (*Gysin maps for vector bundle stacks.*) If $\pi_{\mathfrak{E}} : \mathfrak{E} \rightarrow \mathcal{W}$ is a vector bundle stack over a stack $\mathcal{W}$ which admits a stratification by global quotients, then $\pi_{\mathfrak{E}}^*$ is an isomorphism by [Kre99, Prop. 5.3.2], and we set $0_{\mathfrak{E}}^! := (\pi_{\mathfrak{E}}^*)^!$. Our standing hypotheses include the stratification hypothesis for this reason.

Under our running assumptions, we claim the map (11) exists and is independent of the choice of embedding data. Existence is the content of Lemma 3.5 and independence the content of Lemma 3.6. Once those are established we define the residue map exactly as above in Definition 3.12.

Remark 3.4. Before proceeding to the technical lemmas, we remind the reader that specifying (moving T -equivariant) embedding data is equivalent to specifying a (moving T -equivariant) perfect obstruction theory [Man12, Prop. 3.11], and we will use them interchangeably. If given such a perfect obstruction theory, then the corresponding pair $(j, \mathfrak{E})$ is given by $\mathfrak{E} = h^1/h^0((E^\bullet)^\vee)$ and the composition

$$j : \mathfrak{C}_{\mathcal{Z}/\mathcal{Y}} \hookrightarrow \mathfrak{N}_{\mathcal{Z}/\mathcal{Y}} = h^1/h^0(\tau_{\geq -1}L_{\eta|_{\mathcal{Z}}}^\vee) \xrightarrow{\phi^\vee} \mathfrak{E}.$$

Lemma 3.5 (Existence of embedding data.). *Let $\mathcal{Y}$ be an algebraic stack finite type over k with a T -action and suppose that $\mathcal{Y}^T$ is algebraic. Let $\mathcal{Z} \subseteq \mathcal{Y}^T$ be an open and closed substack which is finite type over k and has the resolution property (every coherent sheaf on $\mathcal{Z}$ is a quotient of finite-rank locally free sheaves). Then there*

exists a T -equivariant perfect obstruction theory

$$\phi : E^\bullet = [E^{-1} \rightarrow E^0] \rightarrow \tau = \tau_{\geq -1} L_\eta|_{\mathcal{Z}}$$

with E^{-1}, E^0 finite-rank moving T -equivariant locally free sheaves on $\mathcal{Z}$. Consequently

$$\mathfrak{E} := h^1/h^0((E^\bullet)^\vee) = [E^1/E^0]$$

is a vector bundle stack over $\mathcal{Z}$, globally presented by finite-rank moving locally free sheaves, and there is a T -equivariant closed embedding of cone stacks

$$j : \mathfrak{E}_{\mathcal{Z}/\mathcal{Y}} \hookrightarrow \mathfrak{E}.$$

In particular, if every connected component of $\mathcal{Y}^T$ is finite type over k and has the resolution property, then the conclusion holds over all of $\mathcal{Y}^T$ componentwise.

Proof. Since η is representable by algebraic spaces it is DM-type (Proposition 2.7(a)), so the intrinsic normal sheaf and cone $\mathfrak{E}_{\mathcal{Z}/\mathcal{Y}} \subseteq \mathfrak{N}_{\mathcal{Z}/\mathcal{Y}}$ are defined and the cone is a subcone of the sheaf [Man12, Prop. 2.33]. Moreover $h^1(L_\eta) = 0$ (by [Man12, Remark 2.3] or [AP24, Proposition 2.24] or [AP24, Prop. 2.24]) so τ has cohomology only in degrees -1 and 0 , and

$$\mathfrak{N}_{\mathcal{Z}/\mathcal{Y}} = h^1/h^0(\tau^\vee).$$

Step 1: coherent representative for τ . We now argue that τ is quasi-isomorphic to a moving complex $[A^{-1} \rightarrow A^0]$ with A^{-1} and A^0 coherent. A quasi-coherent T -equivariant sheaf on $\mathcal{Z}$ is the same as a quasi-coherent sheaf on $\mathcal{Z} \times BT$, so all sheaves in this step are taken to live on the Noetherian algebraic stack $\mathcal{Z} \times BT$.

Note first that h^{-1} and h^0 are coherent. Indeed, coherence can be checked locally, and since Olsson's cotangent complex is compatible with restriction along smooth charts, we can reduce this claim to a morphism of algebraic spaces locally of finite type over BT . Here, $\tau_{\geq -1} L$ is computed by the naive cotangent complex [Stacks, 08RB], whose h^0 and h^{-1} are coherent [Stacks, 0D0U].

Next, because $\mathcal{Z} \times BT$ is Noetherian and τ is bounded with quasi-coherent cohomology, we can find a representative $B^\bullet = [B^{-1} \xrightarrow{d} B^0]$ for τ . Choose $A^0 \subseteq B^0$ to be a coherent subsheaf surjecting onto $h^0(\tau)$ under $B^0 \rightarrow h^0(\tau)$, this must exist by the coherence of $h^0(\tau)$. The submodule $d^{-1}(A^0) \subseteq B^{-1}$ is only quasi-coherent a priori, but the restriction of d to $d^{-1}(A^0)$ gives us the exact sequence

$$0 \rightarrow \ker(d) \rightarrow d^{-1}(A^0) \rightarrow \text{img}(d) \cap A^0 \rightarrow 0,$$

and since the left and right terms are coherent by the coherence of h^0 and h^{-1} , the middle term is as well. Thus $A^{-1} := d^{-1}(A^0)$ is coherent and the inclusion $[A^{-1} \rightarrow A^0] \hookrightarrow [B^{-1} \rightarrow B^0]$ is a quasi-isomorphism. By Lemma 2.11, $L_{\mathcal{Z}/\mathcal{Y}}$ is entirely moving, and hence we can choose $A^\bullet$ to have no fixed part.

Step 2: resolve. We now resolve $A^\bullet$ with finite-rank locally free sheaves, and because we assumed the resolution property on $\mathcal{Z}$, we once again take our sheaves to live on $\mathcal{Z}$ rather than on $\mathcal{Z} \times BT$. To ensure the resolution of $A^\bullet$ is also moving, do this weight by weight: if F is any coherent moving sheaf on $\mathcal{Z}$, then it can be decomposed as $F = F_{\chi_1} \oplus \cdots \oplus F_{\chi_i}$ for finitely many non-zero weights χ_i . Then let $W_i \rightarrow F_{\chi_i}$ be a resolution of F_{χ_i} , and let T act on W_i with pure weight χ_i . Then $\bigoplus_i W_i \rightarrow F$ is equivariant, finite-rank and moving.

We can therefore choose a moving finite-rank locally free surjection $q_0 : E^0 \rightarrow A^0$. Then take

$$P := E^0 \times_{A^0} A^{-1} = \{(e, a) \mid q_0(e) = d(a)\} \subseteq E^0 \oplus A^{-1},$$

which is also moving and coherent, and choose $q_{-1} : E^{-1} \rightarrow P$ to be another moving-finite rank locally free surjection. This gives us a complex $E^\bullet = [E^{-1} \xrightarrow{d_E} E^0]$ and a map $\phi : E^\bullet \rightarrow A^\bullet$ so that $h^0(\phi)$ is an isomorphism and $h^{-1}(\phi)$ is a surjection, hence a T -equivariant moving perfect obstruction theory for η . By Remark 3.4, this is the same as moving T -equivariant embedding data for η . $\square$

We now state the independence lemma.

Lemma 3.6 (Independence of the residue map). *Let $\mathcal{Y}$, $\mathcal{Y}^T$ and $\mathcal{Z} \subseteq \mathcal{Y}^T$ be as in Lemma 3.5. Given any two pairs $(j_1, \mathfrak{E}_1)$ and $(j_2, \mathfrak{E}_2)$ where $j_i : \mathfrak{E}_{\mathcal{Z}/\mathcal{Y}} \hookrightarrow \mathfrak{E}_i$ is a T -equivariant closed embedding and $\mathfrak{E}_i$ is a moving T -equivariant finite rank vector bundle stack for each $i = 1, 2$, then for any class $\alpha \in A_*^T(\mathcal{Y})$,*

$$e_T^{-1}(\mathfrak{E}_1) \cap \eta_1^!(\alpha) = e_T^{-1}(\mathfrak{E}_2) \cap \eta_2^!(\alpha)$$

in $A_^T(\mathcal{Y}^T)_{\text{loc}}$. Here, $\eta_i^!$ is the virtual pullback defined with respect to $(j_i, \mathfrak{E}_i)$.*

Remark 3.7. Before proceeding, two remarks are in order. First, the same specialization map $\sigma : A_*^T(\mathcal{Y}) \rightarrow A_*^T(\mathfrak{E})$ occurs in both $\eta_1^!$ and $\eta_2^!$, hence it suffices to prove that for every $\beta \in A_*^T(\mathfrak{E}_{\mathcal{Z}/\mathcal{Y}})$,

$$(12) \quad e_T(\mathfrak{E}_1)^{-1} \cap 0_{\mathfrak{E}_1}^!((j_1)_*\beta) = e_T(\mathfrak{E}_2)^{-1} \cap 0_{\mathfrak{E}_2}^!((j_2)_*\beta).$$

This removes the stack $\mathcal{Y}$ and the map σ from the discussion entirely.

Second, once the specialization map σ is removed, the cone structure of $\mathfrak{E}$ is no longer relevant, and we can simply take $\mathfrak{E}$ to be any algebraic stack T -equivariant over $\mathcal{Z}$ admitting a T -equivariant closed embedding into $\mathfrak{E}$. This is not a superfluous point; the proof of Lemma 3.6 requires this generality.

For these reasons, we make the following definition.

Definition 3.8. If $\mathfrak{C}$ is an algebraic stack T -equivariant over $\mathcal{Z}$ admitting a T -equivariant closed embedding $j : \mathfrak{C} \hookrightarrow \mathfrak{E}$ into a finite-rank moving vector bundle stack over $\mathcal{Z}$, then for any class $\beta \in A_*^T(\mathfrak{C})$ we define

$$S_{(j, \mathfrak{E})}(\beta) := e_T(\mathfrak{E})^{-1} \cap 0_{\mathfrak{E}}^!(j_*\beta) \in A_*^T(\mathcal{Z})_{\text{loc}}.$$

We first prove Lemma 3.6 in the non-stacky case.

Lemma 3.9 (Residue independence for honest vector bundles). *Let C be a T -stack over $\mathcal{Z}$ admitting T -equivariant closed embeddings $j_1 : C \hookrightarrow E_1$ and $j_2 : C \hookrightarrow E_2$ into two different T -equivariant finite rank moving vector bundles over $\mathcal{Z}$. Then for all $\beta \in A_*^T(C)$,*

$$S_{(j_1, E_1)}(\beta) = S_{(j_2, E_2)}(\beta).$$

We can therefore drop the reference to the embedding data and write $S_C(\beta)$ without ambiguity.

Proof. The strategy is to embed C into $E_1 \oplus E_2$ in two ways: first via

$$i_0 : C \xrightarrow{j_1} E_1 = E_1 \oplus 0 \xhookrightarrow{\iota} E_1 \oplus E_2,$$

and then via the diagonal embedding

$$i_1 : C \hookrightarrow E_1 \oplus E_2, \quad c \mapsto (j_1(c), j_2(c)).$$

First define the interpolation map

$$\varphi : C \times \mathbf{A}^1 \rightarrow E_1 \oplus E_2 \times \mathbf{A}^1; \quad \varphi(c, t) = (j_1(c), t \cdot j_2(c), t),$$

noting that the fibers of φ at $C \times \{0\}$ and $C \times \{1\}$ recover i_0 and i_1 respectively. This map is T -equivariant if we let T act trivially on $\mathbf{A}^1$.

Now let $\text{pr}_C : C \times \mathbf{A}^1 \rightarrow C$ and $\text{pr}_{\mathcal{Z}} : \mathcal{Z} \times \mathbf{A}^1 \rightarrow \mathcal{Z}$ be the projection maps and for $\beta \in A_*^T(C)$ set

$$\begin{aligned} \Theta(\beta) &:= 0_{(E_1 \oplus E_2) \times \mathbf{A}^1}^!(\varphi_* \text{pr}_C^* \beta) \\ &= 0_{(E_1 \oplus E_2) \times \mathbf{A}^1}^!(\varphi_*(\beta \times [\mathbf{A}^1])) \in A_*^T(\mathcal{Z} \times \mathbf{A}^1). \end{aligned}$$

For each t , the inclusion $r_t : \{t\} \hookrightarrow \mathbf{A}^1$ yields Kresch's refined Gysin map $r_t^!$. It acts a map $r_t^! : A_*^T(Y) \rightarrow A_*^T(Y_t)$ for any Y over $\mathbf{A}^1$, it commutes projective pushforward and flat pullback and $r_t^! \circ \text{pr}_C = \text{id}$ on $A_*^T(C)$. Therefore,

$$\begin{aligned} r_t^! \Theta(\beta) &= r_t^! 0_{(E_1 \oplus E_2) \times \mathbf{A}^1}^!(\varphi_*(\beta \times [\mathbf{A}^1])) \\ &= 0_{E_1 \oplus E_2}^!(r_t^! \varphi_*(\beta \times [\mathbf{A}^1])) \\ &= 0_{E_1 \oplus E_2}^!(\varphi_{t*}(\beta)). \end{aligned}$$

Since $r_t^! \Theta(\beta) = (\text{pr}_C)^{-1} \circ \Theta(\beta)$ is independent of t , so is $0_{E_1 \oplus E_2}^!(\varphi_{t*}(\beta))$, and hence

$$0_{E_1 \oplus E_2}^!(i_1\beta) = 0_{E_1 \oplus E_2}^!(\varphi_{1*}(\beta)) = 0_{E_1 \oplus E_2}^!(\varphi_{0*}(\beta)) = 0_{E_1 \oplus E_2}^!(i_0\beta).$$

All that is left is to transport the inverse Euler class through the various morphisms. Let $f : E_1 \oplus E_2 \rightarrow \mathcal{Z}$ be the bundle map and factor it as $f = g \circ h$ where $h : E_1 \oplus E_2 \rightarrow E_1$ is the first projection and $g : E_1 \rightarrow \mathcal{Z}$ is the bundle map of E_1 . This means

$$0_{E_1 \oplus E_2}^! = (f^*)^{-1} = (g^*)^{-1} \circ (h^*)^{-1} = 0_{E_1}^! \circ (h^*)^{-1}.$$

Now remember that we defined $\iota : E_1 \rightarrow E_1 \oplus E_2$, so the self-intersection formula for a class ζ in $A_*^T(E_1)$ reads

$$(h^*)^{-1}(\iota_* \zeta) = e_T(g^* E_2) \cap \zeta,$$

which we apply below to $\zeta = j_{1*} \beta$.

Applying the Whitney sum formula and the projection formula ([Kre99, §3.6]) then yields

$$\begin{aligned} S_{(i_1, E_1 \oplus E_2)}(\beta) &= e_T^{-1}(E_1 \oplus E_2) \cap 0_{E_1 \oplus E_2}^!(i_{1*} \beta) \\ &= e_T^{-1}(E_1) \cdot e_T^{-1}(E_2) \cap 0_{E_1 \oplus E_2}^!(i_{0*} \beta) \\ &= e_T^{-1}(E_1) \cdot e_T^{-1}(E_2) \cap 0_{E_1}^!((h^*)^{-1}(\iota_* j_{1*} \beta)) \\ &= e_T^{-1}(E_1) \cdot e_T^{-1}(E_2) \cap 0_{E_1}^!(e_T(g^* E_2) \cap j_{1*} \beta) \\ &= e_T^{-1}(E_1) \cdot e_T^{-1}(E_2) \cdot e_T(E_2) \cap 0_{E_1}^!(j_{1*} \beta) \\ &= e_T^{-1}(E_1) \cap 0_{E_1}^!(j_{1*} \beta) \\ &= S_{(j_1, E_1)}(\beta). \end{aligned}$$

Swapping the roles of E_1 and E_2 and using the interpolation

$$\varphi'(c, t) = (t \cdot j_1(c), j_2(c), t)$$

gives us $S_{(i_1, E_1 \oplus E_2)}(\beta) = S_{(j_2, E_2)}(\beta)$, and we are done. $\square$

This lemma fails for cone stacks, which is why it isn't a proof of Lemma 3.6 on its own. The following example demonstrates what goes wrong:

Example 3.10. Suppose $(j_i : \mathfrak{C} \rightarrow \mathfrak{E}_i)$ is an embedding of the intrinsic normal cone stack $\mathfrak{C}$ for $\eta : Z \rightarrow X$ into a vector bundle stack $\mathfrak{E}_i$ where $i = 1, 2$. Then it is not necessarily the case that the diagonal map $j = (j_1, j_2) : \mathfrak{C} \rightarrow \mathfrak{E}_1 \oplus \mathfrak{E}_2$ is a closed embedding.

To see this, let $Z = X = \operatorname{Spec} k$, $V = \mathbb{A}^1$ be the rank 1 vector bundle over Z and $\mathfrak{C} = \mathfrak{E}_1 = \mathfrak{E}_2 = [0/V] = BV$. Note that $\mathfrak{C}$ is the intrinsic normal cone of the morphism $\operatorname{Spec} k \rightarrow [\operatorname{Spec} k/V]$. Take $j_i : \mathfrak{C} \rightarrow \mathfrak{E}_i$ to be the identity for both i . The diagonal is then

$$j = (j_1, j_2) : BV \rightarrow BV \oplus BV = B(V \oplus V)$$

induced by the diagonal map $V \rightarrow V \oplus V$. The fiber of a point $\text{Spec } k$ over this diagonal is

$$\text{Spec } k \times_{B(V \oplus V)} BV = V,$$

so the map j has positive dimensional fibers.

On the level of complexes, this can be seen as a failure of h^0 to be an isomorphism (as per the criterion in [BF97, Proposition 2.6]). The two-term truncation of the cotangent complex of $f : \text{Spec } k \rightarrow BV$ is $L_f = [0 \rightarrow k]$ (in degree -1 and 0) and the complex corresponding to $B(V \oplus V)$ is $[0 \rightarrow k^2]$ (dualize and quotient to get $B(V \oplus V)$). The candidate obstruction theory is then determined by the morphism in degree 0 , which is $(a, b) \mapsto a + b$, and this is exactly the map $h^0(\phi)$ as well. This is surjective, but not an isomorphism, hence $\phi^\vee : h^1/h^0(L_f^\vee) \rightarrow h^1/h^0((E_1^\bullet \oplus E_2^\bullet)^\vee)$ fails to be a closed embedding.

We fix this issue by moving from stacky cones and vector bundles to their atlases. Given two choices of stacky embedding data $(j_i, \mathfrak{E}_i)$ with $i = 1, 2$ for the normal cone $\mathfrak{C}_\eta$ where $\mathfrak{E}_i = [E_i^1/E_i^0]$, set

$$C_{E_i^1} = \mathfrak{E}_i \times_{\mathfrak{E}_i} E_i^1$$

and form $j'_i : C_{E_i^1} \hookrightarrow E_i^1$ by pulling j_i back along the atlas $p_i : E_i^1 \rightarrow \mathfrak{E}_i$. This $C_{E_i^1}$ is an honest T -cone over $\mathcal{Z}$; smooth locally on $\mathcal{Z}$ the cone stack $\mathfrak{C}$ is a quotient of a cone by a bundle action, and the atlas pullback recovers that cone. We then show that

$$S_{(j_1, \mathfrak{E}_1)} = S_{(j_2, \mathfrak{E}_2)}$$

by comparing $S_{C_{E_1^1}}$ and $S_{C_{E_2^1}}$. We need one more preliminary result, after which 3.6 is immediate.

Lemma 3.11. *Suppose*

$$0 \rightarrow F \rightarrow D \xrightarrow{p} \mathfrak{C} \rightarrow 0$$

is an exact sequence of moving cone stacks over $\mathcal{Z}$ where F is a finite rank moving vector bundle, D is a moving cone and $\mathfrak{C}$ is a moving cone stack. Then

$$S_C(\beta) = e_T(F) \cap S_D(p^* \beta).$$

Proof. If $j : D \hookrightarrow G$ is an embedding of D into a finite rank moving vector bundle stack over $\mathcal{Z}$, then we get a diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & F & \longrightarrow & G & \xrightarrow{p'} & \mathfrak{C} \longrightarrow 0 \\ & & \uparrow \text{id} & & \uparrow j & & \uparrow j' \\ 0 & \longrightarrow & F & \longrightarrow & D & \xrightarrow{p} & \mathfrak{C} \longrightarrow 0 \end{array}.$$

The properties of Gysin morphisms and flat pullback mean that for any $\beta \in A_*^T(C)$ we have $jp^*(\beta) = p'^*j'_*(\beta)$ and $0_G^! \circ p'^* = 0_E^!$, so

$$\begin{aligned} S_C(\beta) &= e_T(\mathfrak{E})^{-1} \cap 0_E^! j_*(\beta) \\ &= e_T(F) \cdot e_T(F)^{-1} \cdot e_T(\mathfrak{E})^{-1} \cap 0_G^! p'^* j_*(\beta) \\ &= e_T(F) \cdot e_T(G)^{-1} \cap 0_G^! j'_*(p^* \beta) \\ &= e_T(F) \cap S_D(p^* \beta). \end{aligned}$$

□

The independence lemma is now straightforward.

Proof of Lemma 3.6. Let $(j_1, \mathfrak{E}_1)$ and $(j_2, \mathfrak{E}_2)$ be two choices of embedding data for the intrinsic normal cone $\mathfrak{C}_\eta$ of $\eta : \mathcal{Z} \rightarrow \mathcal{Y}$. We wish to show

$$S_{(j_1, \mathfrak{E}_1)}(\beta) = S_{(j_2, \mathfrak{E}_2)}(\beta)$$

for all $\beta \in A_*^T(\mathfrak{C}_\eta)$. Let $\mathfrak{E}_i = [E_i^1/E_i^0]$ be presentations of these vector bundle stacks as global quotients for $i = 1, 2$. Form the following diagram:

$$\begin{array}{ccccc} & & C & & \\ & \swarrow r_1 & & \searrow r_2 & \\ & C_{E_1^1} & \diamond & C_{E_2^1} & \\ & \swarrow j'_1 & & \searrow j'_2 & \\ E_1^1 & & \mathfrak{C}_\eta & & E_2^1 \\ & \swarrow p'_1 & & \searrow p'_2 & \\ & \diamond & & \diamond & \\ & \swarrow p_1 & & \searrow p_2 & \\ & \mathfrak{E}_1 & & \mathfrak{E}_2 & \end{array}$$

For each $i = 1, 2$ the projection $p'_i : C_{E_i^1} \rightarrow \mathfrak{C}_\eta$ is an E_i^0 -torsor over $\mathfrak{C}_\eta$. The fiber product $C = C_{E_1^1} \times_{\mathfrak{C}_\eta} C_{E_2^1}$ is an E_2^0 -torsor over $C_{E_1^1}$ via r_1 and an E_1^0 -torsor over $C_{E_2^1}$ via r_2 , since torsors are compatible with base change.

By Lemma 3.11,

$$(13) \quad S_{(j_i, \mathfrak{E}_i)}(\beta) = e_T(E_i^0) \cap S_{C_{E_i^1}}(p_i'^* \beta)$$

for $i = 1, 2$. Then by repeated application of this formula and Lemma 3.11 we have

$$\begin{aligned} \text{(Equation (13))} \quad S_{(j_1, \mathfrak{E}_1)}(\beta) &= e_T(E_1^0) \cap S_{C_{E_1^1}}(p_1'^* \beta) \\ \text{(Lemma 3.11)} \quad &= e_T(E_1^0) \cap e_T(E_2^0) \cap S_C(r_1^* p_1'^* \beta) \\ \text{(top square commutativity)} \quad &= e_T(E_2^0) \cap e_T(E_1^0) \cap S_C(r_2^* p_2'^* \beta) \\ \text{(Lemma 3.11)} \quad &= e_T(E_2^0) \cap S_{C_{E_2^1}}(p_2'^* \beta) \\ \text{(Equation (13))} \quad &= S_{(j_2, \mathfrak{E}_2)}(\beta), \end{aligned}$$

and we're done. $\square$

Definition 3.12. Let $\mathcal{Y}$ be an algebraic stack of finite type over $\mathrm{Spec} k$ with a T -action and suppose that $\mathcal{Y}^T$ is algebraic. Let $\mathcal{Z} \subseteq \mathcal{Y}^T$ be an open and closed substack which is finite type over k and has the resolution property. We then define the equivariant residue $\mathrm{Res}_{\mathcal{Z}/\mathcal{Y}} : A_*^T(\mathcal{Y}) \rightarrow A_*^T(\mathcal{Z})_{\mathrm{loc}}$ by

$$\mathrm{Res}_{\mathcal{Z}/\mathcal{Y}}(\alpha) = e_T^{-1}(\mathfrak{E}) \cap \eta^!(\alpha),$$

where $\mathfrak{E}$ is any moving equivariant vector bundle stack over $\mathcal{Z}$ admitting an embedding $j : \mathfrak{E}_{\mathcal{Y}^T/\mathcal{Y}} \hookrightarrow \mathfrak{E}$.

4. LOCALIZATION

We now come to the localization formula. Throughout this section $\pi : M \rightarrow \mathcal{Y}$ is a T -equivariant morphism from a Deligne-Mumford stack to an algebraic stack with a T -equivariant perfect obstruction theory $\phi : E \rightarrow L_{M/\mathcal{Y}}$. As in Convention 1.1, we take M to be finite type over k and $\mathcal{Y}$ to be locally of finite type over k . The fixed locus M^T carries the obstruction theory E^f of Lemma 2.12 relative to $\mathcal{Y}^T$, and the residue map $\mathrm{Res}_{\mathcal{Y}^T/\mathcal{Y}}$ transports classes from $\mathcal{Y}$ to a locus $\mathcal{Z} \subseteq \mathcal{Y}^T$.

Remark 4.1 (Admissible substacks). The residue map is built atop Kresch's cycle groups and Manolache's virtual pullback, both of which require $\mathcal{Y}$ to be of finite-type over k , but according to 1.1 we only require it to be locally of finite type. This is because the bases in logarithmic Gromov-Witten theory are most often only locally of finite type, and the fix is standard: replace $\mathcal{Y}$ with a suitable open substack $\mathcal{U} \subseteq \mathcal{Y}$ (see for instance [Ran22]).

Call an open substack $\mathcal{U} \subseteq \mathcal{Y}$ T -stable if the action map sends $T \times \mathcal{U}$ to $\mathcal{U}$ and *admissible* if it is quasi-compact, contains $\pi(M)$ and is T -stable. That such an admissible substack always exists is well known: since M is finite type, $\pi(M)$ is quasi-compact and lies in a quasi-compact open $\mathcal{V} \subseteq \mathcal{Y}$, and taking $\mathcal{U}$ to be the image of $T \times \mathcal{V}$ under the action map means it is also quasi-compact and contains $\pi(M)$.

The choice of admissible $\mathcal{U}$ furthermore does not matter. If $\mathcal{U}_1, \mathcal{U}_2$ are both admissible and $\alpha_i \in A_*^T(\mathcal{U}_i)$ agree on a common admissible $\mathcal{U}_0 \subseteq \mathcal{U}_1 \cap \mathcal{U}_2$ then $\pi_{\mathcal{U}_1}^!(\alpha_1) = \pi_{\mathcal{U}_2}^!(\alpha_2)$ by applying [Man12, Thm. 4.1] to the quotients of the $\mathcal{U}_i$ by T . In particular, a class on $\mathcal{Y}$ has the same virtual pullback for every admissible $\mathcal{U}$.

Remark 4.2. The results of this section therefore apply to bases locally finite over k without modification merely by passing to an admissible $\mathcal{U} \subseteq \mathcal{Y}$. We record some other standing consequences of Convention 1.1 below:

- (1) $\mathcal{Y}$ is locally of finite type over k and by Proposition 2.7 $\eta : \mathcal{Y}^T \rightarrow \mathcal{Y}$ is representable by algebraic spaces and $i : M^T \rightarrow M$ is a closed immersion.

- (2) Since $\mathcal{Y}$ is an algebraic stack of finite type over k with a T -action, affine stabilizers and quasi-compact separated diagonal (Convention 1.1), so is $[\mathcal{Y}/T]$, and furthermore $\mathcal{Y}$, $[\mathcal{Y}/T]$ and every stack representable and separated over them admit stratifications by global quotients. Indeed, the pullback of the diagonal of $[\mathcal{Y}/T]$ to $\mathcal{Y} \times \mathcal{Y}$ is $T \times \mathcal{Y} \rightarrow \mathcal{Y} \times \mathcal{Y}$, $(t, v) \mapsto (tv, v)$. This is a base change of $\Delta_{\mathcal{Y}}$ followed by the projection $T \times \mathcal{Y} \times \mathcal{Y} \rightarrow \mathcal{Y} \times \mathcal{Y}$, so it is quasi-compact and separated. A stabilizer of $[Y/T]$ is an extension of a closed subgroup of T be a stabilizer of $\mathcal{Y}$, hence affine. Finally, stacks of finite type with affine stabilizers are stratified by global quotients [Kre99, Prop. 3.5.9], and a stack representable and separated over one of them has affine stabilizers and quasi-compact separated diagonal by the Transfer Lemma 2.9, hence inherits this property.
- (3) All virtual pullbacks and residues in this section are T -equivariant, seen by applying Manolache's and Kresch's constructions to the quotient stacks $[-/T]$.

Remark 4.3. Three resolutions are needed, each one of a specific object.

- (a) *Embedding data for $\eta|_{\mathcal{Z}}$* : such data is a T -equivariant perfect obstruction theory $[E^{-1} \rightarrow E^0] \rightarrow \tau_{\geq -1} L_{\eta}$ with E^{-1} and E^0 finite rank moving locally free sheaves on $\mathcal{Z}$. This is the only resolution required by Definition 3.12, which defines the residue maps $\text{Res}_{\mathcal{Z}/\mathcal{Y}}$ and $\text{Res}_{\widetilde{M}/M}$.
- (b) *A resolution of E* : this is a T -equivariant quasi-isomorphism $[F^{-1} \rightarrow F^0] \simeq E$ with F^{-1} and F^0 finite rank locally free sheaves on M . This ensures that $h^1/h^0(E^{\vee})$ is a global quotient $[F_1/F_0]$ of vector bundles, required for applying [Man12, Thm. 4.1, Thm. 4.3, Thm. 4.8].
- (c) *A resolution of E^m* : this is a T -equivariant quasi-isomorphism $[N_1^{\vee} \xrightarrow{d^{\vee}} N_0^{\vee}] \simeq E^m$ in degrees -1 and 0 with N_0 and N_1 finite-rank moving locally free sheaves on M^T . It's needed to define N^{vir} , $\check{E}$ and $\text{Res}_{M^T/\widetilde{M}}$. The existence of this resolution follows from (b) by taking the moving part of the restriction of $[F^{-1} \rightarrow F^0]$ to M^T .

The localization theorem relates $(\pi^T)^!$ of the residue $\text{Res}_{\mathcal{Z}/\mathcal{Y}}$ to the residue $\text{Res}_{M^T/M}$ of $\pi^!$, and its proof necessarily passes through the base change $\widetilde{M} = M \times_{\mathcal{Y}} \mathcal{Z}$. The guiding diagram of this section is the following:

$$(14) \quad \begin{array}{ccccc} & & i & & \\ & \nearrow & & \searrow & \\ M^T & \xrightarrow{j} & \widetilde{M} & \xrightarrow{\tilde{i}} & M \\ & \searrow \pi^T & \downarrow \tilde{\pi} & & \downarrow \pi \\ & & \mathcal{Z} & \xrightarrow{\eta} & \mathcal{Y} \end{array}$$

and the proof of the main theorem consists of transporting classes through this diagram via Res maps and virtual pullbacks.

We begin this section by showing M^T is a union of connected components of $\widetilde{M}^T$ with compatible obstruction theories (Section 4.1). Then we prove two intermediate results treating the two halves of the diagram, Proposition 4.5 deals with the rightmost square and Proposition 4.6 treats the leftmost triangle. We conclude the section with the main theorem and show that it recovers the traditional localization theorem after a reparameterization of the torus.

4.1. The fixed locus of the base change. Remember that a point of $\mathcal{Y}^T$ is an object u of $\mathcal{Y}$ with a fixed-point structure α , a compatible family of isomorphisms $\alpha_t : tu \simeq u$, and that if T acts trivially then α is a homomorphism $T \rightarrow \text{Aut}(u)$ (Remark 2.2).

Proposition 4.4. *Let $\pi : M \rightarrow \mathcal{Y}$, $E \rightarrow L_{M/\mathcal{Y}}$, $\mathcal{Z} \subseteq \mathcal{Y}^T$ and $\widetilde{M} = M \times_{\mathcal{Y}} \mathcal{Y}^T$ be as in the section introduction and take the projections $\tilde{i} : \widetilde{M} \rightarrow M$ and $\tilde{\pi} : \widetilde{M} \rightarrow \mathcal{Z}$ as in diagram (14).*

- (a) *$\widetilde{M}$ is a separated Deligne-Mumford stack of finite type with a T -action for which $\tilde{i}$ and $\tilde{\pi}$ are equivariant, and $\tilde{i}$ is representable and separated.*
- (b) *j factors as $M^T \xrightarrow{j'} \widetilde{M}^T \xrightarrow{i'} M$ where j' is an open and closed immersion. Hence j is a closed immersion and M^T is a union of connected components of $\widetilde{M}^T$.*
- (c) *The composite $\tilde{E} := \tilde{i}^*E \rightarrow \tilde{i}^*L_{M/\mathcal{Y}} \rightarrow L_{\widetilde{M}/\mathcal{Z}}$ is a T -equivariant perfect obstruction theory for $\tilde{\pi}$, and j' identifies $(i^*E)^{\text{fix}} \rightarrow L_{M^T/\mathcal{Z}}$ with the restriction of $(i'^*\tilde{E})^{\text{fix}} \rightarrow L_{\widetilde{M}^T/\mathcal{Z}}$, and $L_{M^T/\mathcal{Z}} = L_{M^T/\mathcal{Y}^T}$.*

Proof. *Proof of (a).* The morphism $\tilde{i}$ is the base change of $\eta|_{\mathcal{Z}}$, which is representable and separated by Proposition 2.7(a). This means $\widetilde{M}$ is Deligne-Mumford, and its diagonal is the composite of the closed immersion $\Delta_{\tilde{i}}$ with a base change of Δ_M , which is finite because M is separated. Hence, $\widetilde{M}$ is separated. It is of finite type because both M and $\mathcal{Z}$ are and $\Delta_{\mathcal{Y}}$ is quasi-compact.

Proof of (b). The morphism j is T -equivariant for the trivial action on M^T , so the universal property of the fixed stack [Rom05, Def. 2.3] guarantees it factors as $j = i' \circ j'$ with $j' : M^T \rightarrow \widetilde{M}^T$. By part (a) $\widetilde{M}$ is a separated Deligne-Mumford stack, hence i' is a closed immersion by Proposition 2.7(b). Then since $\tilde{i} \circ j = i$ is a closed immersion and $\tilde{i}$ and i' are separated, j and j' are closed immersions. It remains to show that j' is open.

A point of $\widetilde{M}$ is a point m of M together with a point $z = (u, \alpha)$ of $\mathcal{Z}$, where $\pi(m) = u$ in $\mathcal{Y}$ and α is a fixed-point structure on u . A fixed point of $\widetilde{M}$ is a

such a point together with a fixed-point structure on the whole pair. But $\widetilde{M}^T \simeq M^T \times_{\mathcal{Y}^T} \mathcal{Z}^T$ by Lemma 2.8, so that is a fixed-point structure β on m and a fixed-point structure γ on the point z of $\mathcal{Z}$ which are compatible over $\mathcal{Y}^T$. That is, a collection $((m, \beta), ((u, \alpha), \gamma))$ so that for each $t \in T$, $\pi \circ \beta_t = \gamma_t \circ \alpha_t$. So a fixed point of $\widetilde{M}$ carries two actions of T on the base point u , the one coming from M through β and α coming from the $\mathcal{Z}$ data, and the compatibility over $\mathcal{Y}^T$ asks that these are the same action on u after correcting by γ . From this description it is clear that the points of $\widetilde{M}^T$ corresponding to the image of M^T under j' are those where $\gamma = 1$.

Consider the unit section $e : \mathcal{Z} \rightarrow \mathcal{Z}^T$, $z \mapsto (z, 1)$. Composing with $(\eta|_{\mathcal{Z}})^T$ yields the inclusion $\mathcal{Z} \subseteq \mathcal{Y}^T$, so basechanging e along $\widetilde{M}^T \rightarrow \mathcal{Z}$ is

$$\widetilde{M}^T \times_{\mathcal{Z}^T} \mathcal{Z} = M^T \times_{\mathcal{Y}^T} \mathcal{Z} = (\pi^T)^{-1}(\mathcal{Z}) = M^T \rightarrow \widetilde{M}^T,$$

which is precisely j' . It therefore suffices to show that e is an open immersion. Since e is a monomorphism, it is an open immersion if and only if it is formally étale, which is equivalent to $\tau_{\geq -1} L_e = 0$.

This follows from Lemma 2.11. Since $\eta_{\mathcal{Z}} \circ e = \text{id}$, the transitivity triangle gives $L_e \simeq e^* L_{\eta_{\mathcal{Z}}}[1]$. At a point (z, γ) in $\mathcal{Z}^T$, T act on $L_{\eta_{\mathcal{Z}}}|_z$ through γ , but the image of e consists of all points of the form $(e, \gamma = 1)$, hence the T -action on L_e is trivial. The weight 0 part of $\tau_{\geq -1} L_e$ is trivial by Lemma 2.11, hence $L_e = 0$.

Proof of (c). That $\widetilde{E}$ is a perfect obstruction theory for $\widetilde{\pi}$ is [BF97, Prop. 7.2] and [Man12, Thm. 4.3]. Since j' is an open immersion, the restriction of $i'^* \widetilde{E} \rightarrow L_{\widetilde{M}^T/\mathcal{Z}}$ is $i^* E \rightarrow i^* L_{M/\mathcal{Y}} \rightarrow L_{M^T/\mathcal{Z}}$ by transitivity and functoriality of cotangent complexes for $M^T \rightarrow \widetilde{M} \rightarrow M$ over $\mathcal{Z} \rightarrow \mathcal{Y}$. The weight decompositions correspond because j' is equivariant for trivial actions. Finally $j'^* L_{\widetilde{M}/\mathcal{Z}} = L_{M^T/\mathcal{Z}} = L_{M^T/\mathcal{Y}^T}$ because j' and $\mathcal{Z} \subseteq \mathcal{Y}^T$ are open immersions. $\square$

4.2. Residues, base change and compatible triples. Here we prove the two intermediate results needed to transport a class through diagram (14) from $\mathcal{Y}$ to M^T . Definition 3.12 gives

$$\text{Res}_{\mathcal{Z}/\mathcal{Y}} = e_T(\mathfrak{E})^{-1} \eta^! : A_*^T(\mathcal{Y}) \rightarrow A_*^T(\mathcal{Z})_{\text{loc}}$$

for some choice of embedding data $\mathfrak{E}$, which we fix. The square defining $\widetilde{M}$ in (14) is Cartesian, so $\widetilde{\pi}^*$ of the embedding data is embedding data for $\widetilde{i}$ [Man12, Prop. 2.26]. Write $\widetilde{i}^!$ for the corresponding virtual pullback and define

$$(15) \quad \text{Res}_{\widetilde{M}/M} := e_T(\widetilde{\pi}^* \mathfrak{E})^{-1} \cap \widetilde{i}^! : A_*^T(M) \rightarrow A_*^T(\widetilde{M})_{\text{loc}}.$$

We cannot apply the convention of Section 3.1 directly to $\widetilde{M}$ because it has a non-trivial T action; instead, we take $A_*^T(\widetilde{M})_{\text{loc}}$ to be $S_{\mathcal{Z}}^{-1} A_*^T(\widetilde{M})$ where $S_{\mathcal{Z}}$ is the multiplicative set of Euler classes of finite rank moving bundles on $\mathcal{Z}$ pulled back

to $\widetilde{M}$ via $\widetilde{\pi}$. The map in (15) is therefore not technically an instance of Definition 3.12 since $\widetilde{M}$ is not a fixed locus, but its independence of the choice of embedding data $(j_\eta, \mathfrak{E})$ for $\eta : \mathcal{Z} \rightarrow \mathcal{Y}$ is nonetheless clear from the following proposition and the independence we have already proven (Lemma 3.6).

Proposition 4.5. *Virtual pullback commutes with taking equivariant residues, that is,*

$$\widetilde{\pi}_{E|\widetilde{M}}^! \circ \text{Res}_{\mathcal{Z}/\mathcal{Y}} = \text{Res}_{\widetilde{M}/M} \circ \pi_E^!.$$

In particular, if $\mathcal{Y}$ has a fundamental class,

$$\widetilde{\pi}_{E|\widetilde{M}}^! (\text{Res}_{\mathcal{Z}/\mathcal{Y}}[\mathcal{Y}]) = \text{Res}_{\widetilde{M}/M}([M]^{vir})$$

in $A_*(\widetilde{M})_{\text{loc}}$.

Proof. Since virtual pullback is bivariant, for the left hand side of the desired equality we get

$$\widetilde{\pi}_{E|\widetilde{M}}^! (e_T(\mathfrak{E})^{-1} \cap \eta^!(\alpha)) = \widetilde{\pi}^* e_T(\mathfrak{E})^{-1} \cap (\widetilde{\pi}^! \circ \eta)(\alpha)$$

and on the right side we simply have

$$e_T(\widetilde{\mathfrak{E}})^{-1} \cap \widetilde{i}^! \circ \pi^!.$$

The equality of $\widetilde{\pi}^! \circ \eta^!$ and $\widetilde{i}^! \circ \pi^!$ is precisely [Man12, Theorem 4.3], and $e_T(\widetilde{\mathfrak{E}})^{-1} = e_T(\pi^* \mathfrak{E})^{-1} = \widetilde{\pi}^* (e_T(\mathfrak{E})^{-1})$ follows from the properties of Euler classes.

The particular case is then immediate since $[M]^{vir} = \pi_E^!([\mathcal{Y}])$ by definition when $\mathcal{Y}$ has a virtual fundamental class. $\square$

This takes care of the square

$$\begin{array}{ccc} \widetilde{M} & \xrightarrow{\widetilde{i}} & M \\ \downarrow \widetilde{\pi} & & \downarrow \pi \\ \mathcal{Z} & \xrightarrow{\eta} & \mathcal{Y} \end{array}$$

in diagram (14), all that remains is the leftmost triangle

$$\begin{array}{ccc} M^T & \xrightarrow{j} & \widetilde{M} \\ & \searrow \pi^T & \downarrow \widetilde{\pi} \\ & & \mathcal{Z} \end{array}$$

Functoriality of virtual pullback [Man12, Theorem 4.8] makes this easy. To apply it, we require a compatible triple [Man12, Definition 4.5] for the composition

$$M^T \xrightarrow{j} \widetilde{M} \xrightarrow{\widetilde{\pi}} \mathcal{Z}$$

which consists of a distinguished triangle of perfect obstruction theories

$$\widetilde{\pi}^* E_{\widetilde{M}/\mathcal{Z}}^\bullet \rightarrow E_{M^T/\mathcal{Z}}^\bullet \rightarrow E_{M^T/\widetilde{M}}^\bullet \xrightarrow{+1} \dots$$

with a map to the corresponding distinguished triangle of cotangent complexes. Let E denote the relative perfect obstruction theory for $\pi : M \rightarrow \mathcal{Y}$. The obstruction theory we wish to use for π^T is then $E|_{M^T}^f$ (see Lemma 2.12) and for $\tilde{\pi}$ it is $\tilde{E}$ (see Lemma 4.4), which forces the obstruction theory for j to be the cone of $E|_{M^T}^f \rightarrow \tilde{E}$, which is $E^m := E|_{M^T}^m$ but shifted: $E^m[1]$. This lines up with intuition from the case of a regular embedding $Z \hookrightarrow X$, where the perfect obstruction theory is $L_{Z/X} \simeq N_{Z/X}^\vee[1]$. However, here this doesn't work; $E^m[1]$ is amplitude $[-2, -1]$, so we lose the obstruction bundle when passing to h^1/h^0 and functoriality fails.

There is a clever fix employed by [CKL17] which saves this: we split the candidate $E^m[1]$ by the triangle

$$N_0^\vee[1] \rightarrow E^m[1] \rightarrow N_1^\vee[2] \xrightarrow{+1},$$

keep $N_0^\vee[1]$ as the theory for j and then move the out-of-window part into the obstruction theory for $\tilde{\pi} \circ j$, shifted back down into the appropriate degree:

$$\check{E} := E^f \oplus N_1^\vee[1].$$

This gives a compatible triple factoring the map $\pi^T : M^T \rightarrow \mathcal{Z}$ to which we can apply the functoriality of virtual pullbacks. The resulting formula we get does not involve the perfect obstruction theory we need, but we can bridge the gap easily by factoring π^T as $\pi^T \circ \text{id}_{M^T}$ and finding a second compatible triple. This is the essence of Proposition 4.6.

Fix a T -equivariant quasi-isomorphism

$$(\text{degrees } -1, 0) \quad [N_1^\vee \xrightarrow{d^\vee} N_0^\vee] \simeq E^m$$

with N_0 and N_1 locally free of finite rank moving bundles on M^T and set

$$\check{E} := E^f \oplus N_1^\vee, \quad \check{\phi} := (\phi^f, 0) : \check{E} \rightarrow L_{M^T/\mathcal{Z}}.$$

Proposition 4.6. *In the above notation,*

- (i) $\check{\phi}$ is a T -equivariant relative perfect obstruction theory for $\pi^T : M^T \rightarrow \mathcal{Z}$
- (ii) $(j^*\tilde{E}, \check{E}, N_0^\vee[1])$ is a compatible triple for $M^T \xrightarrow{j} \tilde{M} \xrightarrow{\tilde{\pi}} \mathcal{Z}$.
- (iii) $(E^f, \check{E}, N_1^\vee[1])$ is a compatible triple for $M^T \xrightarrow{\text{id}} M^T \xrightarrow{\pi^T} \mathcal{Z}$.

Proof. *Proof of (i).* $\check{E}$ is the direct sum of two complexes each of perfect amplitude $[-1, 0]$, so it is of perfect amplitude $[-1, 0]$ as well. Since $h^0(N_1^\vee[1]) = 0$ we get $h^0(\check{\phi}) = h^0(\phi^f)$ and $h^{-1}(\check{\phi})$ restricted to the summand $h^{-1}(E^f)$ is $h^{-1}(\phi^f)$, hence $h^0(\check{\phi})$ and $h^{-1}(\check{\phi})$ are an isomorphism and a surjection respectively by 2.12.

Proof of (ii). Let $q : E^m = [N_1^\vee \rightarrow N_0^\vee] \twoheadrightarrow N_1^\vee[1]$ be the projection, with cone $N_0^\vee[1]$. The direct sum of the distinguished triangles

$$E^f \xrightarrow{\text{id}} E^f \rightarrow 0 \quad \text{and} \quad E^m \xrightarrow{q} N_1^\vee[1] \xrightarrow{d^\vee[1]} N_0^\vee[1]$$

is the top row of

$$\begin{array}{ccccccc}
 j^* \tilde{E} = E^f \oplus E^m & \xrightarrow{\text{id} \oplus q} & \check{E} & \xrightarrow{(0, d^\vee[1])} & N_0^\vee[1] & \xrightarrow{+1} & \\
 \downarrow j^* \tilde{\phi} & & \downarrow \check{\phi} & & \downarrow \mathfrak{f} & & \\
 j^* L_{\tilde{M}/\mathcal{Z}} & \longrightarrow & L_{M^T/\mathcal{Z}} & \longrightarrow & L_{M^T/\tilde{M}} & \xrightarrow{+1} &
 \end{array}$$

whose bottom row is the transitivity triangle of cotangent complexes. The left square commutes because $j^* \tilde{\phi} = (\phi^f, 0)$ and $\mathfrak{f}$ exists by axiom (TR3) of distinguished triangles completing the two rows to a morphism of triangles. Fix any such map $\mathfrak{f}$. We are then in exactly the situation of [Man12, Construction 3.13] and the diagram chase there shows $\mathfrak{f}$ is a perfect obstruction theory.

Proof of (iii). The diagram for the second compatible triple is

$$\begin{array}{ccccccc}
 E^f & \xrightarrow{(\text{id}, 0)} & \check{E} = E^f \oplus N_1^\vee[1] & \xrightarrow{(0, \text{id})} & N_1^\vee[1] & \xrightarrow{0} & \\
 \downarrow \phi^f & & \downarrow \check{\phi} & & \downarrow 0 & & \\
 L_{M^T/\mathcal{Y}^T} & \xrightarrow{\text{id}} & L_{M^T/\mathcal{Y}^T} & \longrightarrow & L_{\text{id}_{M^T}} = 0 & \xrightarrow{+1} &
 \end{array}$$

The top row is split and hence is a distinguished triangle. The map $0 : N_1^\vee[1] \rightarrow 0$ is a perfect obstruction theory because h^0 vanishes on both sides and $h^{-1}(N_1^\vee[1]) \rightarrow 0$ is surjective. $\square$

Proposition 4.7. *For all $\beta \in A_*^T(\mathcal{Z})$, we have*

$$(16) \quad e_T(N_1) \cap (\pi^T)_{E^f}^!(\beta) = j^!(\tilde{\pi}_{\check{E}}^!(\beta)) \in A_*^T(M^T),$$

and additionally

$$(17) \quad e_T(N^{\text{vir}}) \cap \text{Res}_{M^T/\tilde{M}}(\tilde{\pi}_{\check{E}}^!(\alpha)) = (\pi^T)_{E^f}^!(\alpha).$$

Note that this holds before localization.

Proof. Applying functoriality of virtual pullbacks [Man12, Theorem 4.8] to the first compatible triple of Proposition 4.6 yields

$$(\pi^T)_{\check{E}}^!(\alpha) = j^!(\tilde{\pi}_{\check{E}}^!(\alpha)).$$

Using the fact that

$$\text{id}_{N_1^\vee[1]}^! = 0_{N_1}^! \circ 0_{N_1^*} = e_T(N_1) \cap (-).$$

and applying functoriality to the second compatible triple yields

$$(\pi^T)_{\check{E}}^!(\alpha) = \text{id}_{N_1^\vee[1]}^!((\pi^T)_{E^f}^!(\alpha)) = e_T(N_1) \cap (\pi^T)_{E^f}^!(\alpha).$$

Equating these two gives the formula (16).

To get the (17) we compute $\text{Res}_{M^T/\widetilde{M}}$ with the embedding data (i, N_0) where $i : C_{M^T/\widetilde{M}} \rightarrow N_0$ is the embedding of the normal cone of $M^T \subseteq \widetilde{M}$ into N_0 . We get

$$\text{Res}_{M^T/\widetilde{M}}(\gamma) = e_T(N_0)^{-1} \cap j^!(\gamma)$$

which, combined with (17), is the desired result. $\square$

4.3. The Localization Theorem. Here is the relevant diagram one more time for convenience:

$$\begin{array}{ccccc} & & i & & \\ & \nearrow & & \searrow & \\ M^T & \xrightarrow{j} & \widetilde{M} & \xrightarrow{\tilde{i}} & M \\ & \searrow \pi^T & \downarrow \tilde{\pi} & & \downarrow \pi \\ & & \mathcal{Z} & \xrightarrow{\eta} & \mathcal{Y} \end{array}$$

We define the *virtual normal bundle* as the class $N^{\text{vir}} = [N_0] - [N_1] = [(E^m)^\vee]$, with

$$e_T(N^{\text{vir}}) = e_T(N_0)/e_T(N_1) \in A_T^*(M^T)_{\text{loc}}.$$

It is the class of the moving tangent bundle minus the moving obstruction bundle, matching the definition from [GP99]. We can now state the main theorem.

Theorem 4.8. *Under the standing hypotheses at the beginning of this section and the consequences of Remark 4.2, for every $\beta \in A_*^T(\mathcal{Y})$,*

$$(18) \quad (\pi^T)_{E^f}^! (\text{Res}_{\mathcal{Z}/\mathcal{Y}}(\beta)) = e_T(N^{\text{vir}}) \cap \text{Res}_{M^T/M} (\pi_E^!(\beta)) \quad \text{in } A_*^T(M^T)_{\text{loc}}.$$

More precisely, for every $\alpha \in A_^T(\mathcal{Z})$,*

$$(19) \quad (\pi^T)_{E^f}^! (\alpha) = e_T(N^{\text{vir}}) \cap \text{Res}_{M^T/\widetilde{M}} (\tilde{\pi}_E^!(\alpha))$$

and (18) is (19) with $\alpha = \text{Res}_{\mathcal{Z}/\mathcal{Y}}(\beta)$. Consequently, if $\mathcal{Y}$ is pure-dimensional, then we have a virtual fundamental class $[M]^{\text{vir}} := \pi_E^![\mathcal{Y}]$ and

$$(20) \quad (\pi^T)_{E^f}^! (\alpha) = e_T(N^{\text{vir}}) \cap \text{Res}_{M^T/\widetilde{M}} ([M]^{\text{vir}})$$

for $\alpha = \text{Res}_{\mathcal{Z}/\mathcal{Y}}([\mathcal{Y}])$.

Proof. This is merely a composition of Proposition 4.5 and Proposition 4.7. $\square$

Remark 4.9. Two remarks are in order.

- (1) When $\mathcal{Y}$ is a point and hence $\mathcal{Z} = \mathcal{Y}^T = \mathcal{Y}$, one has $\text{Res}_{\mathcal{Z}/\mathcal{Y}} = \text{id}$, $\widetilde{M} = M$, $\text{Res}_{M^T/M} = e_T(N_0)^{-1} \cap i^!$, and (20) then reads

$$[M^T]^{\text{vir}} = e_T(N^{\text{vir}}) \cap \text{Res}_{M^T/M}([M]^{\text{vir}}).$$

which matches the form of [GP99]. This does not directly recover the classical localization formula of [GP99] as written, because the Romagny fixed stack M^T differs from the fixed stack used by Graber-Pandharipande — see Corollary 4.12 below.

- (2) Because $\mathcal{Z}$ is not pure dimensional in general, there is no virtual fundamental class $[M^T]$ relative to the obstruction theory E^f . It's replacement in (20) is the class $(\pi^T)^!(\alpha)$ for $\alpha = \text{Res}_{\mathcal{Z}/\mathcal{Y}}([\mathcal{Y}])$. While it is not a virtual fundamental class in the Behrend-Fantechi sense – in particular it lives in $A_*^T(M^T)_{\text{loc}}$ rather than $A_*^T(M^T)$ – it doesn't depend on any choices made and it plays the role of the virtual fundamental class in the localization formula.

4.4. Comparison to the absolute case. We conclude this section by comparing the relative case to the classical, absolute case in [GP99]. The classical localization formula is stated for a Deligne-Mumford stack M with a $\mathbb{G}_m$ -equivariant closed embedding into a nonsingular one, Y , and for the fixed substack $M^f = M \cap Y^f$ where Y^f is the zero locus of the vector field of the action. The induced action on M^f , in the sense of Romagny, only becomes trivial after replacing T with a finite cover (see [Ara+25, §0.4] for instance). To recover the classical virtual localization theorem, therefore, we need an intrinsic replacement for M^f and a reparameterization of T , i.e. a surjective homomorphism $\rho : T' \rightarrow T$ of algebraic tori with finite kernel.

The intrinsic replacement for M^f is the stabilizer fixed stack.

Definition 4.10. ([Ara+25, Def. A.43]) Let M be a quasi-DM stack with a T -action. The *stabilizer fixed stack* M^{sT} is the stack over M whose A -valued points, for an M -scheme $x : A \rightarrow M$, are the closed subgroups of $\text{Aut}_{[M/T]}(x)$ that are affine and smooth over A with connected fibers of dimension r .

Here the stabilizer of x is a subgroup of T_A and $\text{Aut}_{[M/T]}(x)$ is an extension of that subgroup by $\text{Aut}_M(x)$. If $F \subseteq [M/T]$ is the locus of points with stabilizer of maximal dimension constructed by Edidin and Rydh, then $M^{sT} = F \times_{[M/T]} M$ whenever $M^{sT} \neq \emptyset$ [Ara+25, Remark A.44].

Proposition 4.11. *Let k be a field of characteristic 0 and let M be a separated Deligne-Mumford stack of finite type over k with T -action. There is a reparameterization $\rho : T' \rightarrow T$ such that, for the action of T' through ρ ,*

$$M^{T'} \simeq M^{rhT} \simeq M^{sT}$$

over M , where $M^{rhT} := \lim_{\rho} M^{hT'}$ is the reparameterization homotopy fixed stack [Ara+25, Def. A.37]. The same holds for every reparameterization factoring through ρ .

Proof. This is a combination of statements from Aranha et. al.'s appendix: [Ara+25, Prop. A.38, Thm A.45, Thm A.46]. $\square$

This allows us to recover the classical virtual localization theorem.

Corollary 4.12. *Suppose that $\mathcal{Y} = \operatorname{Spec} k$. Then there exists a reparameterization $\rho : T' \rightarrow T$ such that, in $A_*^T(M)_{\operatorname{loc}}$*

$$(21) \quad [M]^{\operatorname{vir}} = \iota_* \sum_i \frac{[X_i]^{\operatorname{vir}}}{e_{T'}(N_i^{\operatorname{vir}})},$$

where $M^{T'} = \coprod_i X_i$ is the decomposition of $M^{T'}$ into connected components, $[X_i]^{\operatorname{vir}}$ is the virtual class of the obstruction theory $(E|_{X_i})^f$ of Lemma 2.12 and N_i^{vir} is the moving part of $E^\vee|_{X_i}$. This recovers the formula of [GP99] for the action of T' through ρ .

Proof. Let ρ be as in Proposition 4.11, so that $M^{T'} = M^{rhT}$ contains every geometric point of M whose T -stabilizer is all of T . Then $\iota : M^{T'} \rightarrow M$ is a closed immersion by Proposition 2.7 and $M^{T'}$ has finitely many connected components since M is finite type over k . With $\mathcal{Y} = \mathcal{U} = \mathcal{Z} = \operatorname{Spec} k$ we get $\widetilde{M} = M$ and $\eta = \operatorname{id}$, so restricting (20) to X_i gives us

$$[X_i]^{\operatorname{vir}} = e_{T'}(N_i^{\operatorname{vir}}) \cap \operatorname{Res}_{X_i/M} [M]^{\operatorname{vir}}.$$

The product $X_i \times_M X_j$ is X_i when $i = j$ and is empty otherwise, and since virtual pullback commutes with proper pushforward,

$$\operatorname{Res}_{X_i/M} \iota_{i*} \beta = e_{T'}(N_0)^{-1} (e_{T'}(N_0) \cap \beta) = \beta$$

by the self intersection formula. Since $M^{T'}$ contains all points with full T' -stabilizer, $\iota_* = \bigoplus \iota_{i*}$ is an isomorphism after inverting the non zero characters of T' [Ara+25, Thm. D], so $\operatorname{Res}_{M^{T'}/M}$ is its inverse and applying ι_* to (20) in Theorem 4.8 gives (21). $\square$

5. PUNCTURED STABLE MAPS

We now verify that the machinery developed in sections 2-4 applies to our intended application to the setting of punctured logarithmic maps. The main reference is [Abr+25], but everything applies to the non-punctured setting of [Abr+20] as well.

The outcome is Proposition 5.4. Under two mild hypotheses on the Artin fan and the type, the bases $\mathfrak{M}$ and $\mathfrak{M}^{\operatorname{ev}}$ satisfy Convention 1.1, the only part of which is not already in [Abr+25] being that they have affine stabilizers and quasi-compact separated diagonal (Proposition 5.3). The existence of the resolutions of Remark 4.3 in this setting is not established in the present draft; see Remark 4.3.

Setting. This is the setting of [Abr+25], whose results we use throughout. We also assume that the Artin fan $\mathcal{A}_X$ below has a Zariski open cover by Artin cones $[V_\sigma/T_\sigma]$ and that the graph G of the type τ fixed below is not a single vertex of genus one without legs.

We fix a log smooth morphism $p : X \rightarrow B$ of fs log schemes of finite type over k to match the assumptions at the beginning of [Abr+25]. This means we also assume X is connected, that B is separated of finite type over k and that $X \rightarrow B$ fits into a diagram

$$\begin{array}{ccc} X & \xrightarrow{q_0} & \mathcal{A}_X \\ \downarrow p & & \downarrow \\ B & \longrightarrow & \mathcal{A}_B \end{array}$$

with strict horizontal arrows, $\mathcal{A}_B$ the Artin fan of B , and $\mathcal{A}_X$ an Artin fan representable over Log . The torus T acts on X via automorphisms of log schemes over B , and trivially on B ; we write $a : T \times X \rightarrow X$ for the action. The *relative Artin fan* is $\mathcal{Y} := B \times_{\mathcal{A}_B} \mathcal{A}_X$, and $q = (p, q_0) : X \rightarrow \mathcal{Y}$ is the induced strict smooth morphism.

We fix a global decorated type $\tau = (G, \mathbf{g}, \boldsymbol{\sigma}, \bar{\mathbf{u}}, \mathbf{A})$ [Abr+25, Def. 2.44] and let $M := \mathcal{M}(X/B, \tau)$ be the stack of τ -marked basic stable punctured maps [Abr+25, Def. 3.8]. It is a Deligne-Mumford stack, and we will need that it is finite type and separated over B . We stick to our convention of denoting Deligne-Mumford stacks by standard letters, and we reserve $\mathcal{M}$ for denoting log structures.

Finally, there are two different bases we consider. The first is $\mathfrak{M} := \mathfrak{M}(\mathcal{Y}/B, \tau)$ the stack of τ -marked basic stable punctured maps to the Artin fan $\mathcal{Y}$, and the map $\pi : \mathcal{M} \rightarrow \mathfrak{M}$ which composes a punctured map with q makes $\mathcal{M}$ a virtually smooth stack over $\mathfrak{M}$ with a perfect obstruction theory $E \rightarrow L_{\mathcal{M}/\mathfrak{M}}$ [Abr+25, Prop. 4.2]. The second base is the *evaluation base*, defined for a subset $\mathbf{S} \subseteq E(G) \cup L(G)$ of edges and legs by the fiber product

$$\mathfrak{M}^{\text{ev}} := \mathfrak{M} \times_{\mathcal{Y}^{\mathbf{S}}} X^{\mathbf{S}}$$

of the evaluation map at the sections $s \in \mathbf{S}$ with $q^{\mathbf{S}}$ [Abr+25, Def. 5.14]. Here we mean $\mathcal{Y}^{\mathbf{S}} = \prod_{s \in \mathbf{S}} \mathcal{Y}$ and $X^{\mathbf{S}} = \prod_{s \in \mathbf{S}} X$. We write $\varepsilon : \mathcal{M} \rightarrow \mathfrak{M}^{\text{ev}}$ for the induced morphism, and it carries its own perfect obstruction theory $E^{\text{ev}} \rightarrow L_{\mathcal{M}/\mathfrak{M}^{\text{ev}}}$.

Remark 5.1. The *weakly marked stacks* $\mathcal{M}'(X/B, \tau)$ and $\mathfrak{M}'(\mathcal{Y}/B, \tau)$ are another variant of these moduli spaces appearing in the punctured stable map theory, and they contain the marked moduli spaces as closed substacks. Every argument in this section applies to them unchanged.

In order to apply Theorem 4.8 to $\mathfrak{M}$ and $\mathfrak{M}^{\text{ev}}$ we must check the properties in Convention 1.1 are met and that the resolutions listed in Remark 4.3 exist. The only pieces of which are not contained in [Abr+25] are that the bases have quasi-compact, separated diagonal and affine stabilizers and the verification of the existence of resolution (a) in Remark 4.3.

We handle the diagonals of the bases by comparing $\mathfrak{M}$ to the stack of underlying nodal curves, to which the results of Hall and Rydh on Hom-stacks apply. The

comparison rests on the fact that a basic log structure contributes no automorphisms to a nodal curve once its underlying data has been fixed.

Let $\mathbf{M}(G, \mathbf{g})$ be the stack of $(G, \mathbf{g})$ -marked pre-stable curves [Abr+25, §3.1.1] with universal curve $\mathcal{C}$ and let

$$\mathcal{H} = \underline{\mathrm{Hom}}_{\mathbf{M}(G, \mathbf{g}) \times B}(\mathcal{C} \times B, \mathcal{Y} \times \mathbf{M}(G, \mathbf{g}))$$

be the stack of pairs consisting of a marked curve over a B -scheme and a morphism of underlying stacks from the curve to $\mathcal{Y}$ over B . Forgetting all log structures defines a functor $F : \mathfrak{M} \rightarrow \mathcal{H}$.

Lemma 5.2. *For every object y of $\mathfrak{M}$ over a scheme, $\mathrm{Aut}(y) \rightarrow \mathrm{Aut}(Fy)$ is injective; hence F is representable by algebraic spaces.*

Proof. Let $y = (C^\circ/W, \mathbf{p}, f)$ be a point of $\mathfrak{M}$ over a scheme $\underline{W}$. An automorphism of y is a triple (φ, θ, ψ) where φ is an automorphism of the underlying marked curve over $\underline{W} \rightarrow B$, θ is a 2-isomorphism $\theta : \underline{f} \implies \underline{f} \circ \varphi$ of maps to $\mathcal{X}$ and $\psi = (\psi_W, \psi_C, \psi_{C^\circ})$ is a collection of automorphisms of the log structures on W , C and C° over φ compatible with $\mathcal{M}_W \rightarrow \mathcal{M}_C \subseteq \mathcal{M}_{C^\circ}$. Since F is the forgetful functor throwing away log structures, the image of (φ, θ, ψ) in $\mathrm{Aut}(Fy)$ is (φ, θ) . Since a morphism of algebraic stacks which is injective on automorphism groups is representable by algebraic spaces [Stacks, Tag 04YY], it suffices to show that when $\varphi = \mathrm{id}$ and $\theta = \mathrm{id}$, $\psi = \mathrm{id}$ as well.

We reduce to the scheme-target case, whose punctured form is [Abr+25, Prop. 2.37] and is proven in [GS13, Prop. 1.25]. Set $X' = (\underline{C}, f^*\mathcal{M}_X)$, an fs log scheme over B through $\underline{C} \rightarrow \underline{W} \rightarrow B$ and $\mathcal{M}_B \rightarrow f^*\mathcal{M}_X$, and $f' = (\mathrm{id}_{\underline{C}}, f^\flat) : C^\circ \rightarrow X'$ so that f is simply f' followed by the strict morphism $X' \rightarrow \mathcal{X}$. Since $\theta = \mathrm{id}$, the log structure $f^*\mathcal{M}_X$ is fixed and ψ is an automorphism of the punctured map $(C^\circ/W, \mathbf{p}, f')$ to X' over the identity of $\underline{C}^\circ$. This is a punctured map and inherits the pre-stability and basicness from $f^*\mathcal{M}_X$ on C and $f^\flat$. The log structure of X' is Zariski since f lifts to a chart V_σ Zariski-locally on $\underline{C}$, and X' is simple in the sense of [Abr+25, Def. 2.1] because $\Sigma(X') \rightarrow \Sigma(\mathcal{X})$ is injective on cones and $\mathcal{X}$ is simple. Hence $\psi = \mathrm{id}$ by [Abr+25, Prop. 2.37]. $\square$

Proposition 5.3. *Assume that $\mathcal{A}_X$ has a Zariski open cover by Artin cones $\mathcal{A}_\sigma = [V_\sigma/T_\sigma]$ and that G is not a single vertex of genus one without legs. Then $\mathfrak{M}$ and $\mathfrak{M}^{ev}$ have affine stabilizers and quasi-compact separated diagonal.*

Proof. First we show that $\Delta_{X/B}$ is separated. By [Stacks, Tag 0CL0], $\Delta_{X/B}$ is separated if and only if the inertia $I_X \rightarrow \mathcal{X}$ is, and the latter can be checked on an open cover. The map $\mathcal{X} \rightarrow \mathcal{A}_X$ is the base change of $B \rightarrow \mathcal{A}_B$ and is hence representable, so $I_X \rightarrow I_{\mathcal{A}_X} \times_{\mathcal{A}_X} \mathcal{X}$ is a monomorphism and it suffices that $I_{\mathcal{A}_X} \rightarrow \mathcal{A}_X$

is separated. Over the open substack $\mathcal{A}_\sigma \subseteq \mathcal{A}_X$ this is the inertia of $[V_\sigma/T_\sigma]$, whose pullback to V_σ is the subgroup $\{(h, v) \mid hv = v\}$ of $T_\sigma \times V_\sigma$. This subgroup is closed, hence affine and hence separated, so $\mathcal{X} \rightarrow B$ has separated diagonal.

Next we show that $\mathcal{H}$ is algebraic locally of finite type over k with affine stabilizers and quasi-compact separated diagonal. The relative form of this comes from [HP23, Thm. 1.2], which says that $\mathcal{H} \rightarrow \mathbf{M}(G, \mathbf{g}) \times B$ is algebraic, locally of finite presentation and quasi-separated with affine stabilizers and separated diagonal. The diagonal of $\mathbf{M}(G, \mathbf{g})$ is separated and of finite presentation [Stacks, Tag 0DSQ] and B is separated, so composing implies $\Delta_{\mathcal{H}}$ is quasi-compact and separated. For the stabilizers, the automorphism group of a geometric point of $\mathbf{M}(G, \mathbf{g})$ is a closed subgroup of $\prod_{v \in V(G)} \text{Aut}(C_v)$ and a factor is non-affine exactly when C_v is a smooth genus-one curve without special points, the case we've excluded. A stabilizer of $\mathcal{H}$ is an extension of a closed subgroup of this affine group, hence affine.

Factorization of $\Delta_{\mathfrak{M}}$: In what follows we factor $\Delta_{\mathfrak{M}}$ as

$$\Delta_{\mathfrak{M}} : \mathfrak{M} \xrightarrow{\Delta_F} \mathfrak{M} \times_{\mathcal{H}} \mathfrak{M} \xrightarrow{\rho} \mathfrak{M} \times_k \mathfrak{M}$$

where ρ is the base change of Δ_H along $F \times_k \times F$.

Separated diagonal: The diagonal Δ_F is a monomorphism and hence separated by Lemma 5.2 and $\Delta_{\mathcal{H}}$ is separated by the above argument. Note that this is the argument of [GS13, Prop 2.5].

Quasi-compact diagonal: Quasi-compactness is stable under base change and composition, and since the base change of $\Delta_{\mathcal{H}}$ is quasi compact by what we have already shown, it suffices to show Δ_F is quasi-compact. That is, since $\mathfrak{M}$ is locally of finite type, that $\text{Isom}_{\mathfrak{M}}(y_1, y_2) \rightarrow \text{Isom}_{\mathcal{H}}(Fy_1, Fy_2)$ is quasi-compact for $y_i = (C_i^\circ/W_i, \mathbf{p}, f_i)$ over a Noetherian scheme S_0 . An isomorphism in $\mathfrak{M}$ consists of an isomorphism of punctured curves, a 2-isomorphism of underlying maps and the compatibility of the two morphisms of log structures f_i^b . Let R' be the algebraic space over S_0 of triples $(\varphi, \theta, \lambda)$ so that the morphism $\alpha : R' \rightarrow \text{Isom}_{\mathcal{H}}(Fy_1, Fy_2)$ simply forgets the data of λ . Because $\mathbf{M}_B(G, \mathbf{g}) = \text{Log}_{\mathbf{M}(G, \mathbf{g}) \times B}$, the diagonal of Log is of finite presentation [Ols03, Thm. 3.2] and the stack $\mathbf{M}_B(G, \mathbf{g})$ of punctured curves is representable and quasi-separated over $\mathbf{M}_B(G, \mathbf{g})$ [Abr+25, Prop. 3.3], the isomorphisms of log curves over B lifting φ form a space of finite presentation and hence α is quasi-compact and locally of finite type. Thus R' is locally of finite type over S_0 , and $C_{i, R'} = C_i \times_{S_0} R'$ is locally Noetherian.

This makes $\text{Isom}_{\mathfrak{M}}(y_1, y_2) \subseteq R'$ the locus over which the two morphisms of log structures

$$u = f_1^b, v = \lambda^b \circ \varphi^* f_2^b \circ \theta^* : N := f_1^* \mathcal{M}_{\mathcal{X}} \rightarrow \mathcal{M}_{C_1^\circ}$$

on $C_{1,R'}$ agree, and this locus is locally closed. Indeed, the locus $U \subseteq C_{1,R'}$ where $\bar{u} = \bar{v}$ is open by, for instance, the proof of [Wis16, Lemma 2.2.1], and on it $u = \varepsilon v$ for a unique $\varepsilon : \overline{N}^{\text{gp}} \rightarrow \mathcal{O}_U^\times$ by integrality of $\mathcal{M}_{C_1^\circ}$ and the condition that $\varepsilon = 1$ is closed by [Wis16, Lemma 2.2.2]. Hence $\text{Isom}_{\mathcal{M}}(y_1, y_2) \rightarrow R'$ is an immersion into a locally Noetherian space, hence quasi-compact.

Affine stabilizers: For a geometric point y of $\mathfrak{M}$, $\text{Aut}(y)$ is of finite type over k since $\Delta_{\mathfrak{M}}$ is quasi-compact and locally of finite type over k . Furthermore $\text{Aut}(y) \rightarrow \text{Aut}_{\mathcal{H}}(Fy)$ is a monomorphism into an affine group. A monomorphism of group schemes of finite type over a field is a closed immersion, so $\text{Aut}(y)$ is also affine.

The evaluation base: $\mathfrak{M}^{\text{ev}} \rightarrow \mathfrak{M}$ is a base change of $q^{\mathbf{S}} : \prod X \rightarrow \prod \mathcal{X}$ which is representable by algebraic spaces and separated. The factorization of the diagonal of $\Delta_{\mathfrak{M}}$ used above then transfers to $\mathfrak{M}^{\text{ev}}$. $\square$

The verification that Theorem 4.8 applies to $\mathfrak{M}$ and $\mathfrak{M}^{\text{ev}}$ is now a checklist.

Proposition 5.4. *Assume that $\mathcal{A}_X$ has a Zariski open cover by Artin cones, that G is not a single vertex of genus one without legs and that M is separated and of finite type over B . Then $\mathfrak{M}$ and $\mathfrak{M}^{\text{ev}}$ meet Convention 1.1.*

Proof. *Local finiteness:* Both bases are locally of finite type over k .

Torus action and equivariance: The torus T acts trivially on $\mathfrak{M}$ and through the factor $X^{\mathbf{S}}$ on $\mathfrak{M}^{\text{ev}}$ so that π and ε are equivariant.

The perfect obstruction theories for the bases is given in [Abr+25, Eq. (4.4)]. It is built from the 2-commutative diagram formed by $\mathcal{C} \rightarrow M$, the universal curve over $\mathfrak{M}$ and the targets $X \rightarrow \mathcal{Y}$. Their construction is carried out for general log stacks and is compatible with base change. It is the pullback of the diagram formed by $[\mathcal{C}/T] \rightarrow [M/T]$, the universal curve over $\mathfrak{M} \times BT$ and $[X/T] \rightarrow \mathcal{Y} \times BT$, which yields the equivariance of the obstruction theories. The same applies to the obstruction theory of $\mathfrak{M}^{\text{ev}} \rightarrow \mathfrak{M}$.

Diagonal and stabilizers: Both bases have quasi-compact separated diagonal and affine stabilizers by Proposition 5.3. $\square$

Remark 5.5. To apply Theorem 4.8 to $\pi : M \rightarrow \mathfrak{M}$ or $\varepsilon : M \rightarrow \mathfrak{M}^{\text{ev}}$ one also needs the resolutions (a) and (b) of Remark 4.3: a two-term resolution of $\tau_{\geq -1} L_\eta$ on a chosen $\mathcal{Z} \subseteq \mathfrak{M}^T$ by moving locally free sheaves, and a two-term resolution of E by locally free sheaves on M . We have not yet determined in what generality (a) exists for the bases of punctured maps, and this is the piece missing from the present draft. This verification, and examples, will appear in a later version.

REFERENCES

- [AB84] M. F. Atiyah and R. Bott. “The Moment Map and Equivariant Cohomology”. *Topology* 23.1 (1984), pp. 1–28.
- [Abr+20] D. Abramovich, Q. Chen, M. Gross, and B. Siebert. “Decomposition of Degenerate Gromov–Witten Invariants”. *Compositio Mathematica* 156.10 (2020), pp. 2020–2075.
- [Abr+25] D. Abramovich, Q. Chen, M. Gross, and B. Siebert. *Punctured Logarithmic Maps*. 1st ed. Vol. 15. Memoirs of the European Mathematical Society. EMS Press, 2025.
- [AP24] D. Aranha and P. Pstrągowski. “The Intrinsic Normal Cone for Artin Stacks”. *Annales de l’Institut Fourier* 74.1 (2024), pp. 71–120.
- [Ara+25] D. Aranha, A. A. Khan, A. Latyntsev, H. Park, and C. Ravi. *Virtual Localization Revisited*. 2025. arXiv: 2207.01652 [math].
- [BF97] K. Behrend and B. Fantechi. “The Intrinsic Normal Cone”. *Inventiones mathematicae* 128.1 (1 1997), pp. 45–88.
- [CKL17] H.-L. Chang, Y.-H. Kiem, and J. Li. “Torus Localization and Wall Crossing for Cosection Localized Virtual Cycles”. *Advances in Mathematics* 308 (2017), pp. 964–986.
- [EG98] D. Edidin and W. Graham. “Localization in Equivariant Intersection Theory and the Bott Residue Formula”. *American Journal of Mathematics* 120.3 (1998), pp. 619–636. JSTOR: 25098611.
- [GP99] T. Graber and R. Pandharipande. “Localization of Virtual Classes”. *Inventiones mathematicae* 135.2 (2 1999), pp. 487–518.
- [GS13] M. Gross and B. Siebert. “Logarithmic Gromov-Witten Invariants”. *Journal of the American Mathematical Society* (2013).
- [GV05] T. Graber and R. Vakil. “Relative Virtual Localization and Vanishing of Tautological Classes on Moduli Spaces of Curves”. *Duke Mathematical Journal* 130.1 (2005), pp. 1–37.
- [HP23] D. Halpern-Leistner and A. Preygel. “Mapping Stacks and Categorical Notions of Properness”. *Compositio Mathematica* 159.3 (2023), pp. 530–589.
- [Kon95] M. Kontsevich. “Enumeration of rational curves via torus actions”. In: *The moduli space of curves (Texel Island, 1994)*. Vol. 129. Progr. Math. Birkhäuser Boston, Boston, MA, 1995, pp. 335–368.
- [Kre99] A. Kresch. “Cycle Groups for Artin Stacks”. *Inventiones mathematicae* 138.3 (1999), pp. 495–536.
- [Man12] C. Manolache. “Virtual pull-backs”. *J. Algebraic Geom.* 21.2 (2012), pp. 201–245.
- [MR19] S. Molcho and E. Routis. “Localization for Logarithmic Stable Maps”. *Transactions of the American Mathematical Society, Series B* 6.3 (2019), pp. 80–113.
- [Ols03] M. Olsson. “Logarithmic Geometry and Algebraic Stacks”. *Annales Scientifiques de l’École Normale Supérieure* 36.5 (2003), pp. 747–791.

- [Ran22] D. Ranganathan. “Logarithmic Gromov–Witten Theory with Expansions”. *Algebraic Geometry* (2022), pp. 714–761.
- [Rom05] M. Romagny. “Group Actions on Stacks and Applications”. *Michigan Mathematical Journal* 53.1 (2005).
- [Rom22] M. Romagny. *Algebraicity and Smoothness of Fixed Point Stacks*. 2022. arXiv: 2205.11114 [math].
- [SG26] B. Siebert and v. M. Garrel. “From wall structures to closed mirror symmetry”. Preliminary draft. 2026.
- [SGA3-II] M. Demazure and A. Grothendieck, eds. *Schémas en groupes II: Groupes de type multiplicatif, et structure des schémas en groupes généraux*. Lecture Notes in Mathematics 152. Séminaire de Géométrie Algébrique du Bois Marie 1962–64 (SGA 3). With the collaboration of M. Artin, J.-E. Bertin, P. Gabriel, M. Raynaud and J.-P. Serre. New edition available at <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. Berlin–New York: Springer-Verlag, 1970.
- [Sil99] J. H. Silverman. *Advanced Topics in the Arithmetic of Elliptic Curves*. Corr. 2. print. Graduate Texts in Mathematics 151. New York Heidelberg Berlin: Springer, 1999.
- [Stacks] T. Stacks Project Authors. *Stacks Project*. <https://stacks.math.columbia.edu>. 2018.
- [STV15] T. Schürg, B. Toën, and G. Vezzosi. “Derived Algebraic Geometry, Determinants of Perfect Complexes, and Applications to Obstruction Theories for Maps and Complexes”. *Journal für die reine und angewandte Mathematik (Crelles Journal)* 2015.702 (2015), pp. 1–40.
- [Wis16] J. Wise. “Moduli of Morphisms of Logarithmic Schemes”. *Algebra & Number Theory* 10.4 (2016), pp. 695–735.